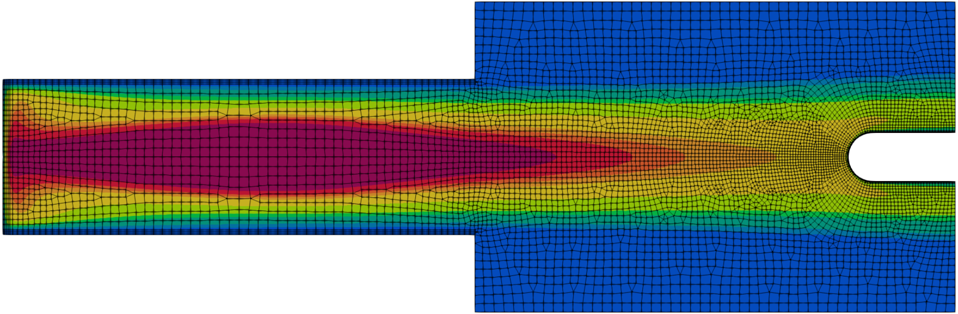
A Hybridized Discontinuous Galerkin Solver for Inductively Coupled Plasma

Nicolas Corthouts



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Introduction

Etats de la matière: eau à pression atmosphérique

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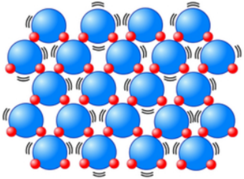
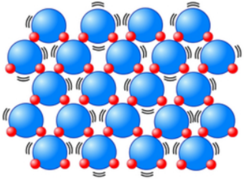
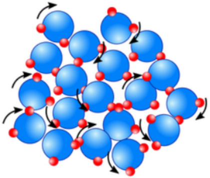
Solide



Etats de la matière: eau à pression atmosphérique

Solide | **Liquide**





Etats de la matière: eau à pression atmosphérique

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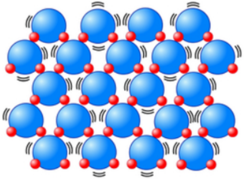
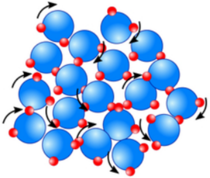
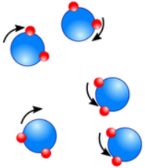


Liquide

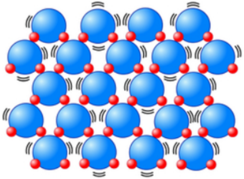


Gaz

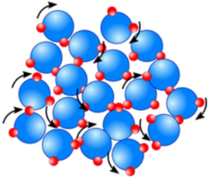


Etats de la matière: eau à pression atmosphérique

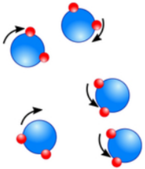
Solide



Liquide

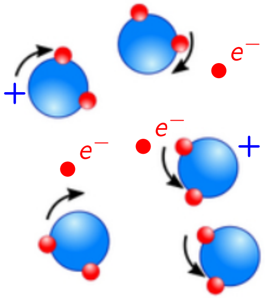


Gaz



Que se passe-t-il si on chauffe encore plus?

Plasma: le quatrième état de la matière

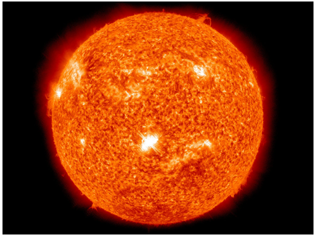


Un plasma est un gaz quasi-neutre composé de particules chargées et neutres démontrant un comportement collectif.[1]

[1] F. F. Chen. Introduction to Plasma Physics and Controlled Fusion. Ed. by Springer International Publisher. 2016

Plasma dans la vie de tous les jours

Les plasmas composent 90% de l'univers visible.




[2]

Fusion nucléaire, médecine, métallurgie, lasers, création de microprocesseurs, ...

[2] Courtesy of Wikipedia.

Plasma en réentrée atmosphérique

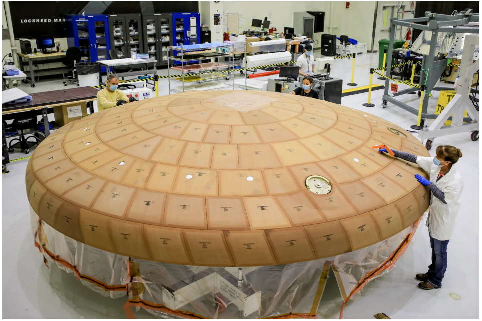
[3]

Vitesse de réentrée $\simeq 10\,\mathrm{km\,s^{-1}}$: choc transformant l'air en plasma.

[3] Coutesy of NASA.

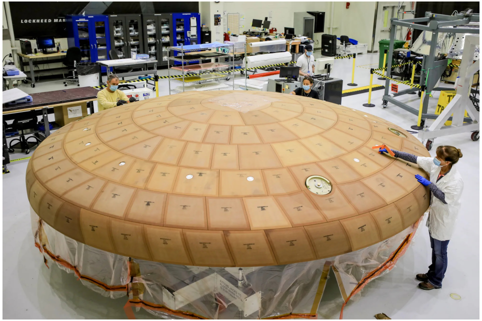
Sytème de protection thermique et destruction des déchets spatiaux

Sytème de protection thermique et destruction des déchets spatiaux

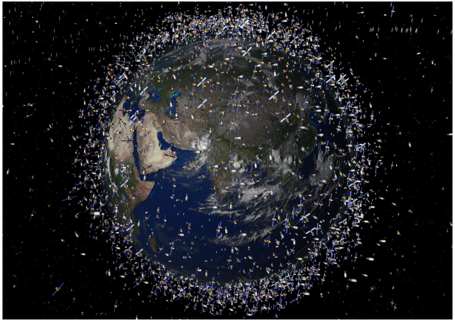


TPS

Sytème de protection thermique et destruction des déchets spatiaux

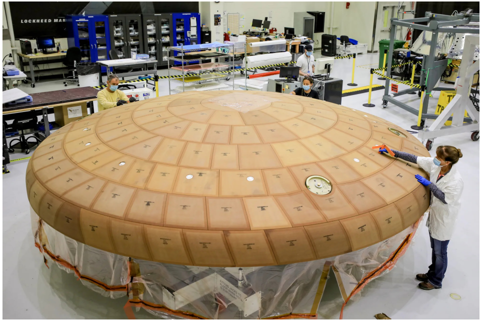


TPS

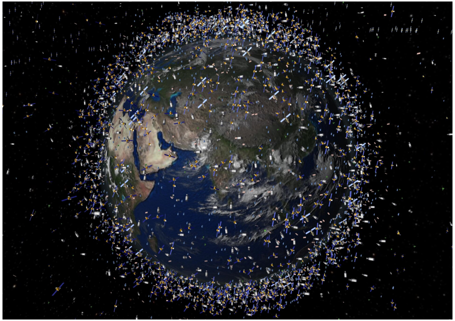


Déchets spatiaux

Sytème de protection thermique et destruction des déchets spatiaux



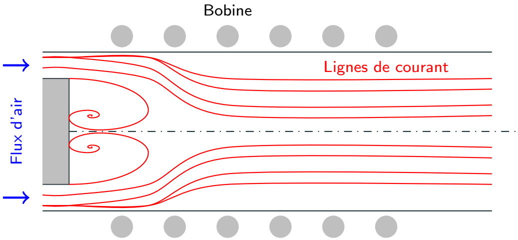
TPS



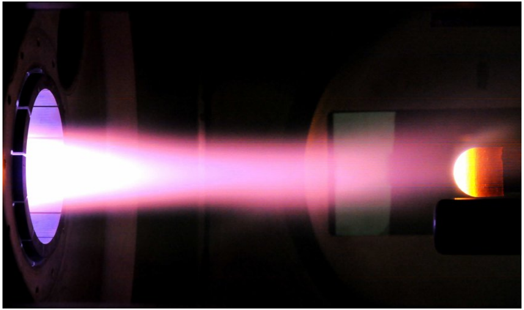
Déchets spatiaux

Développement de dispositifs expérimentaux reproduisant les plasmas de réentrée atmosphérique.

Plasma à induction



Torche



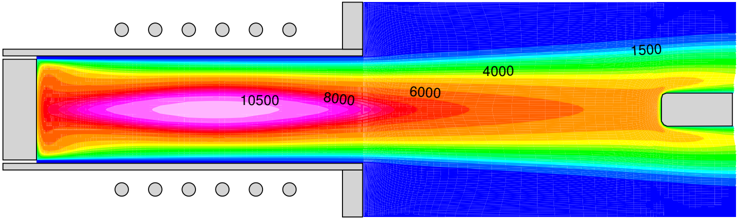
Chambre

Examples: Plasmatron (VKI), Plasmatron X (Illinois), IPG (Russie), ...

Basé sur le principe de **transfer de chaleur local** (video).

Plasma à induction: besoin de plus

Des solvers numériques ont été développés afin de préparer au mieux les expériences dans les torches à induction.



[4]

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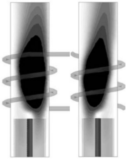
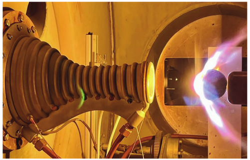
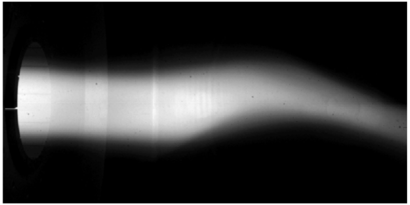
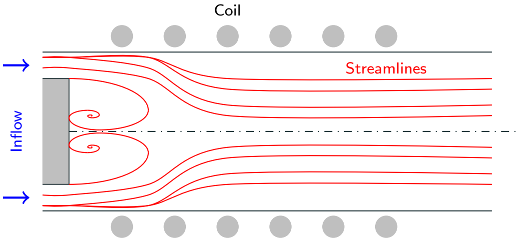
[4] Courtesy of Thierry Magin.

Objectif de la thèse

On propose un nouveau type de solver numérique pour les plasmas afin de représenter plus de situations physiques.

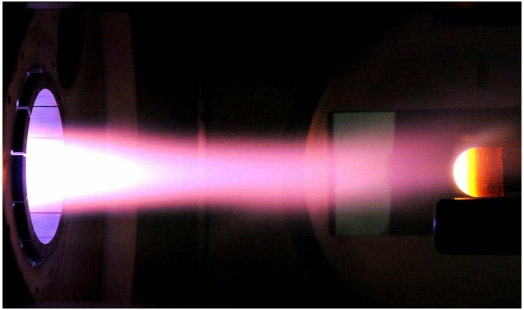




Torch

T 100 000 K, $p \in$ [1000 Pa; 100 000 Pa].

Air, argon, carbon dioxide.

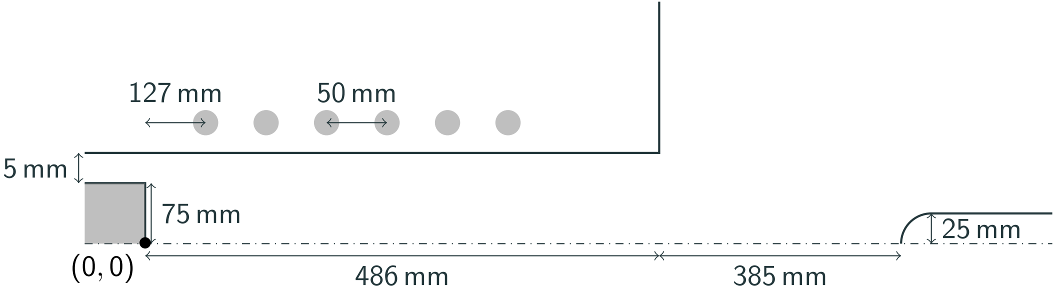


Chamber

Re 100, *Ma* 0.01.

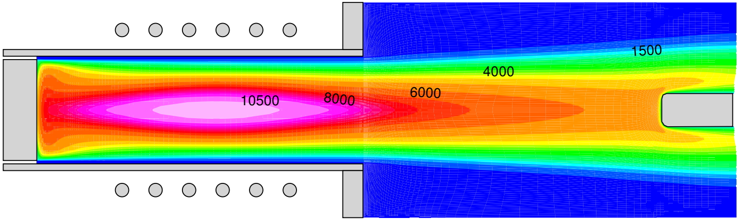
At LTE & non equilibrium.

ICP : geometry



Large temperature gradients!

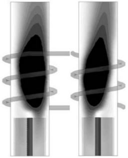
ICP : existing solvers

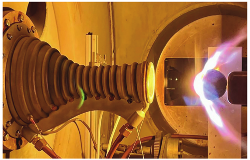


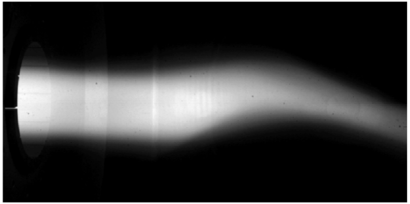
- Axisymmetric.
- Subsonic.
- Steady-state.
- Local thermodynamic equilibrium.
- No-radiation.

- Relatively slow convergence.
- Requires a constant attention during convergence processes.

ICP : what is missing?

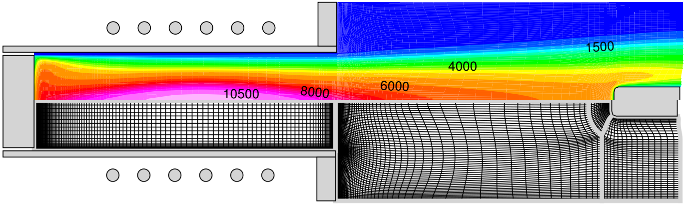






Can we offer a framework able to support future ICP simulations requirements?

Finite volume solver limitations.



FV require a highly refined mesh in regions of large solution gradient.

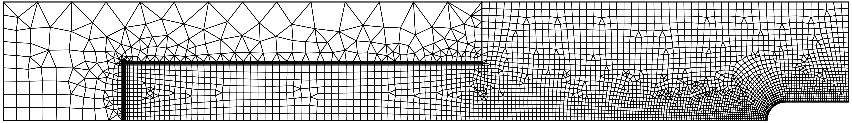
FV can be very dissipative, smoothing small scale phenomena.

Solution procedure makes convergence slow and tedious...

Thesis objective

Developing a discontinuous Galerkin solver of inductively coupled plasmas.

- Solution is not constant anymore on every element.
- Much greater mesh flexibility and (hopefully) less mesh elements.
- We hope to make a more robust solver.
- More precise (less dissipative) solver than finite volumes.



Research questions

With the objective of producing a new high order solver for inductively coupled plasma, we will try to answer the following questions:

Q1 **In addition of being precise, can a high-order solver be robust for inductively coupled plasma applications?**

Q2 **Is the developed solver user-friendly?**

Q3 **Can the new solver be easily adapted to the new experiments performed in ICP facilities?**

Model for inductively coupled plasma

Plasma governing equations : the Boltzmann equation

$$\mathcal{D}_i(f_{i,I}) = \partial_t f_{i,I} + \mathbf{c}_i \cdot \nabla f_{i,I} + \frac{\mathbf{F}_i}{m_i} \cdot \partial_{\mathbf{c}_i} f_{i,I} = \mathcal{S}_i + \mathcal{C}_i$$

$f_i(\mathbf{x}, \mathbf{c})$: probability density function.

$\mathcal{S}_i$: scattering collision operator.

$\mathcal{C}_i$: chemically reacting collision operator.

Provided $Kn \ll 1$ (continuous medium) + conservation of mass, momentum and energy, it is possible to simplify this system.

Plasma collective behaviour

A plasma is a quasi-neutral gas composed of charged and neutral particles exhibiting a <u>collective</u> behaviour (through collisions = particles interaction).

Short ranged

Direct contact (local & binary)

Long ranged

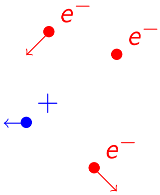
Electric force (distance and collective)

For large scales ($> 1\,\mu\mathrm{m}$ in our case), $q \simeq 0$ due to electric force.

Cold plasmas and thermal equilibrium

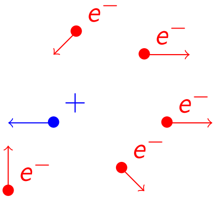
"Cold plasmas" store energy in the free electrons. It is then transmitted during collisions to the ions et neutrals.

Innefficient collisions $(T_e \gg T_h)$



Thermal non-equilibrium

Efficient collisions $(T_e \simeq T_h)$



Thermal equilibrium

Plasmas are in thermal equilibrium in this work.

Chemical equilibrium state

Collisions may lead to chemical reactions.

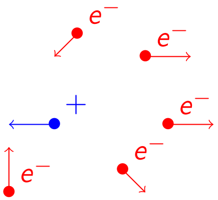
If τ_{chem} and τ_{hydro} are characteristic chemical reaction and flow time respectively:

$\tau_{chem} \gg \tau_{hydro}$	$\tau_{chem} \simeq \tau_{hydro}$	$\tau_{chem} \ll \tau_{hydro}$
Chemical equilibrium	Chemical non-equilibrium	Frozen flow

Plasmas are in chemical equilibrium in this work.

Hypothesis on the plasma

- **Collision-dominated and thermal plasma**



$$\boxed{T_e \simeq T_h}$$

Hypothesis on the plasma

- **Collision-dominated and thermal plasma**
- **Quasi-neutrality**

Characteristic length $\gg$ Debye length so that $q \simeq 0$.

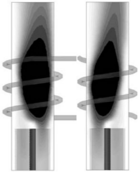
Hypothesis on the plasma

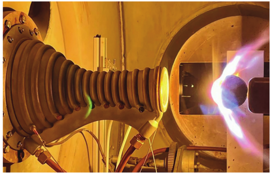
- **Collision-dominated and thermal plasma**
- **Quasi-neutrality**
- **No displacement current**

- Electrons return fast to equilibrium when perturbed by ELM waves.
- ELM wavelength $\gg$ plasma length scale.

Hypothesis on the plasma

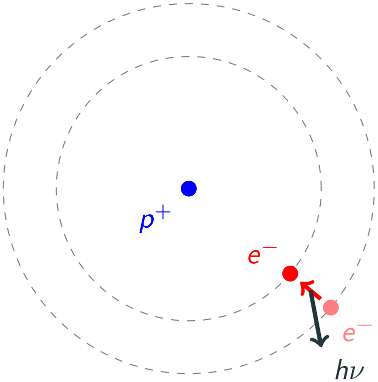
- Collision-dominated and thermal plasma
- Quasi-neutrality
- No displacement current
- Axisymmetry





Hypothesis on the plasma

- **Collision-dominated and thermal plasma**
- **Quasi-neutrality**
- **No displacement current**
- **Axisymmetry**
- **Optically thick plasma**



Hypothesis on the plasma

- **Collision-dominated and thermal plasma**
- **Quasi-neutrality**
- **No displacement current**
- **Axisymmetry**
- **Optically thick plasma**
- **Local thermodynamic equilibrium**

Thermal + chemical equilibrium = LTE

Hypothesis on the plasma

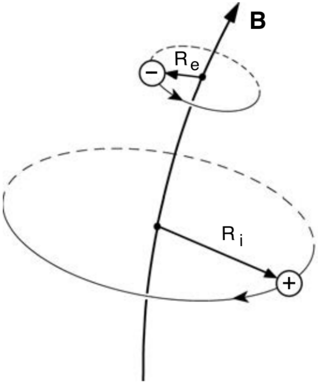
- **Collision-dominated and thermal plasma**
- **Quasi-neutrality**
- **No displacement current**
- **Axisymmetry**
- **Optically thick plasma**
- **Local thermodynamic equilibrium**
- **No elemental de-mixing**

Elemental de-mixing = diffusion of elements for LTE.

Results closer to non-equilibrium.

Hypothesis on the plasma

- Collision-dominated and thermal plasma
- Quasi-neutrality
- No displacement current
- Axisymmetry
- Optically thick plasma
- Local thermodynamic equilibrium
- No elemental de-mixing
- Unmagnetized plasma



Hypothesis on the plasma

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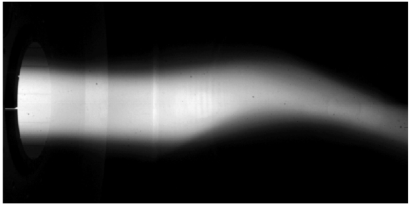
- **Unmagnetized plasma**
- **Steady-state**



Equations averaged over one period of the induction current.

Hypothesis on the plasma

- **Collision-dominated and thermal plasma**
- **Quasi-neutrality**
- **No displacement current**
- **Axisymmetry**
- **Optically thick plasma**
- **Local thermodynamic equilibrium**
- **No elemental de-mixing**
- **Unmagnetized plasma**
- **Steady-state**

The electric field:

- It is ambipolar ($j_z = j_r = 0$).
- The coils are thin parallels wires surrounding the facility.
- Axisymmetric
- $E_{tot} = E_C + E_P$
- We use phasor notation.

Equations governing the plasma

Navier-Stokes equations

$$
\begin{aligned}
&\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0 \\
&\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v}\mathbf{v}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \mathbf{F}_L \\
&\partial_t \left(\rho e + \frac{1}{2}\rho ||\mathbf{v}||^2 \right) + \nabla \cdot \left(\rho e \mathbf{v} + \frac{1}{2}\rho ||\mathbf{v}||^2 \mathbf{v} + p \mathbf{v} \right) = \nabla \cdot (\boldsymbol{\tau}\mathbf{v}) + \nabla \cdot \mathbf{q} + P_J
\end{aligned}
$$

Equations governing the plasma

Navier-Stokes equations

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$$
$$\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v}\mathbf{v}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \mathbf{F}_L$$
$$\partial_t \left(\rho e + \frac{1}{2}\rho ||\mathbf{v}||^2\right) + \nabla \cdot \left(\rho e \mathbf{v} + \frac{1}{2}\rho ||\mathbf{v}||^2 \mathbf{v} + p\mathbf{v}\right) = \nabla \cdot (\boldsymbol{\tau}\mathbf{v}) + \nabla \cdot \mathbf{q} + P_J$$

$$\boldsymbol{\tau} = \eta \left(\nabla \mathbf{v} + \nabla \mathbf{v}^T\right) - \frac{2}{3}\eta \nabla \cdot \mathbf{v}$$
$$\mathbf{q} = -k \nabla T$$

$$F_z^L = \frac{\sigma_e}{4\pi f}\left[E_I^{Im}\partial_z E_I^{Re} - E_I^{Re}\partial_z E_I^{Im}\right]$$
$$F_r^L = \frac{\sigma_e}{4\pi f}\left[E_I^{Im}\frac{1}{r}\partial_r(rE_I^{Re}) - E_I^{Re}\frac{1}{r}\partial_r(rE_I^{Im})\right]$$
$$P^J = \frac{\sigma_e}{2}\left[(E_I^{Im})^2 + (E_I^{Re})^2\right]$$

Electric field equation

$$\partial^2_{zz} E_P + \frac{1}{r}\partial_r(r\partial_r E_P) - \frac{E_P}{r^2} = i2\pi f \mu_0 \sigma_e (E_C + E_P)$$

Electric field equation

$$\boxed{\partial^2_{zz} E_P + \frac{1}{r}\partial_r(r\partial_r E_P) - \frac{E_P}{r^2} = i2\pi f \mu_0 \sigma_e (E_C + E_P)}$$

μ_0 : magnetic permeability

σ_e : electron electric conductivity

f : induction frequency

Electric field equation

$$\boxed{\partial^2_{zz} E_P + \frac{1}{r}\partial_r(r\partial_r E_P) - \frac{E_P}{r^2} = i2\pi f \mu_0 \sigma_e (E_C + E_P)}$$

$$E_C = \sum_{l=1}^{N_{coil}} i f \mu_0 I_C \sqrt{\frac{r_0}{r}} \left[2\frac{E_2(k_l)}{k_l} - E_1(k_l)\left(\frac{2}{k_l} - k_l\right)\right]$$

$E_1(k)$, $E_2(k)$ elliptic integral of the first and second kind.

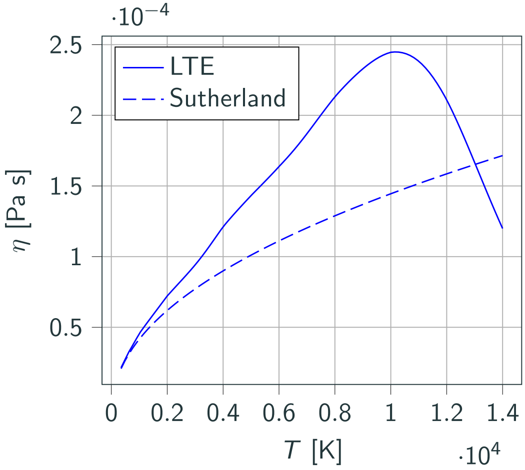
Transport properties : Mutation++

Mutation++

Multicomponent Thermodynamic And Transport properties for IONized gases in C++

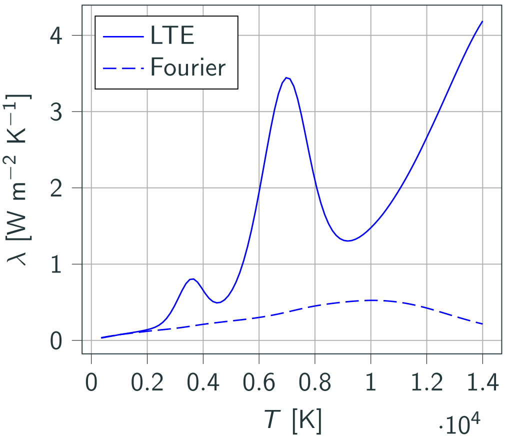
Uses a Galerkin approximate solution of the linearized Boltzmann equation for computing the transport coefficients (σ_e, η, λ).

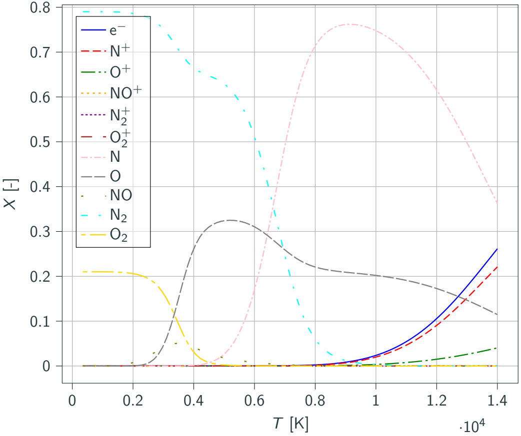
Transport properties : viscosity



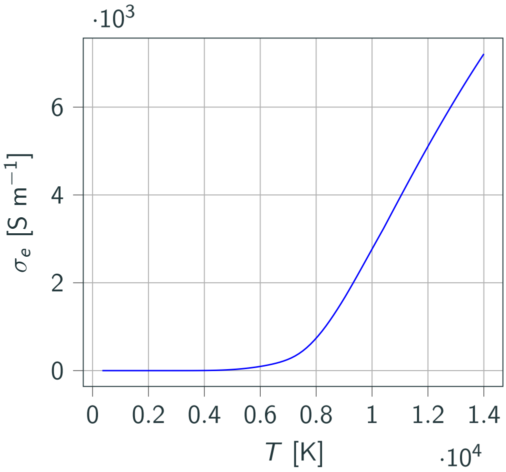
$$\eta_i \propto \frac{\sqrt{T_i}}{\sigma_i}.$$

Transport properties : heat conductivity





Transport properties : electric conductivity



Coupling of the systems

Navier-Stokes

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v}\mathbf{v}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \mathbf{F}_L$$

$$\partial_t \left(\rho e + \frac{1}{2}\rho ||\mathbf{v}||^2 \right) + \nabla \cdot \left(\rho e \mathbf{v} + \frac{1}{2}\rho ||\mathbf{v}||^2 \mathbf{v} + p\mathbf{v} \right) = \nabla \cdot (\boldsymbol{\tau}\mathbf{v}) + \nabla \cdot \mathbf{q} + P_J$$

Navier-Stokes

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v} \mathbf{v}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \mathbf{F}_L$$

$$\partial_t \left(\rho e + \frac{1}{2} \rho ||\mathbf{v}||^2 \right) + \nabla \cdot \left(\rho e \mathbf{v} + \frac{1}{2} \rho ||\mathbf{v}||^2 \mathbf{v} + p \mathbf{v} \right) = \nabla \cdot (\boldsymbol{\tau} \mathbf{v}) + \nabla \cdot \mathbf{q} + P_J$$

Maxwell

$$\partial_{zz}^2 E_P + \frac{1}{r} \partial_r (r \partial_r E_P) - \frac{E_P}{r^2} = i 2 \pi f \mu_0 \sigma_e (E_C + E_P)$$

Coupling of the systems

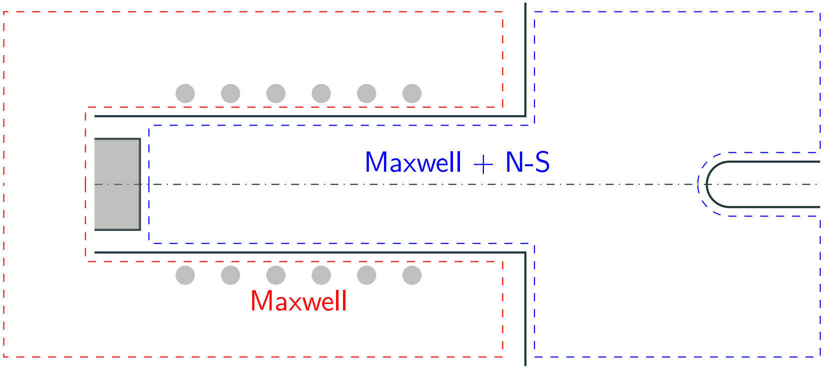
Navier-Stokes

$$
\begin{aligned}
&\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0 \\
&\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v}\mathbf{v}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \mathbf{F}_L \\
&\partial_t \left(\rho e + \frac{1}{2}\rho ||\mathbf{v}||^2 \right) + \nabla \cdot \left(\rho e \mathbf{v} + \frac{1}{2}\rho ||\mathbf{v}||^2 \mathbf{v} + p\mathbf{v} \right) = \nabla \cdot (\boldsymbol{\tau}\mathbf{v}) + \nabla \cdot \mathbf{q} + P_J
\end{aligned}
$$

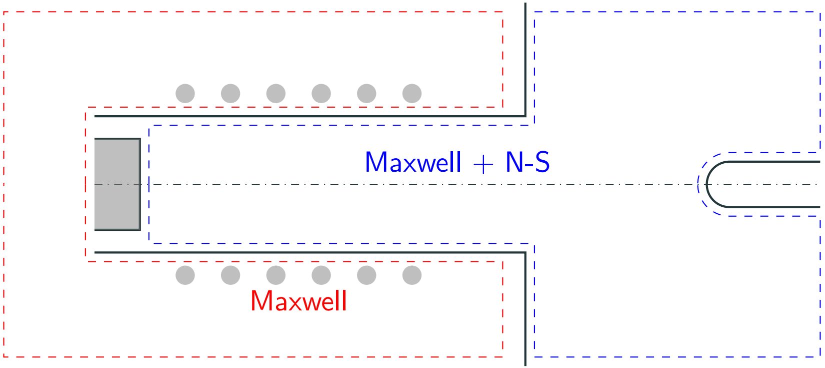
Maxwell

$$
\partial^2_{zz} E_P + \frac{1}{r}\partial_r (r \partial_r E_P) - \frac{E_P}{r^2} = i2\pi f \mu_0 \sigma_e (E_C + E_P)
$$

Computational domain

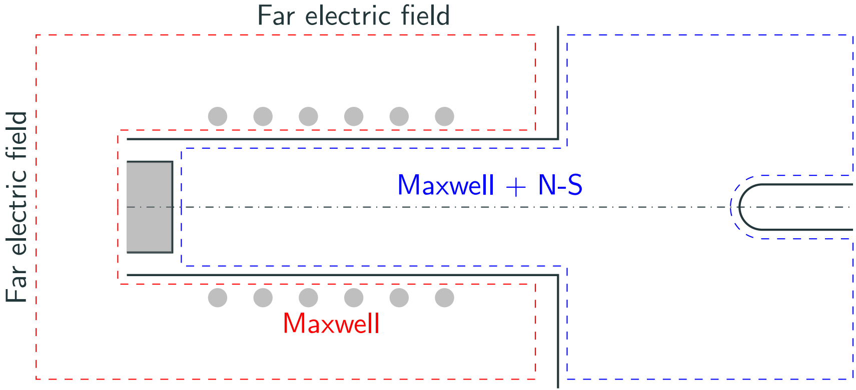


Computational domain



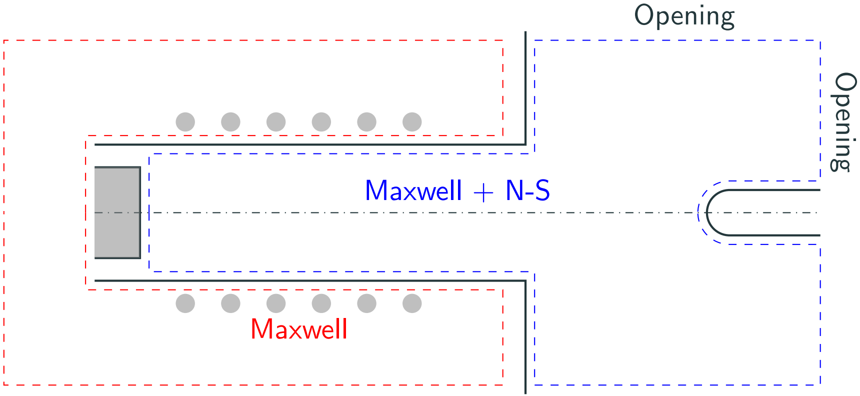
Symmetry axis $\partial_r p = 0$, $\partial_r u_z = 0$, $u_r = 0$, $\partial_r T = 0$ and $E_P = 0$.

Computational domain



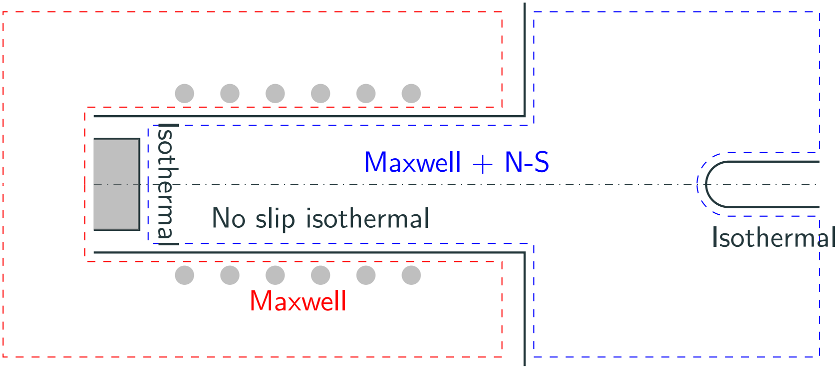
Vanishing far electric field $E_P = 0$.

Computational domain



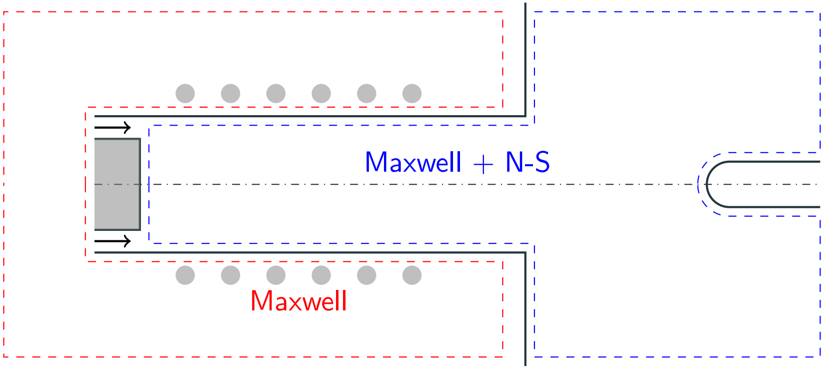
Openings $p = p_0$

Computational domain



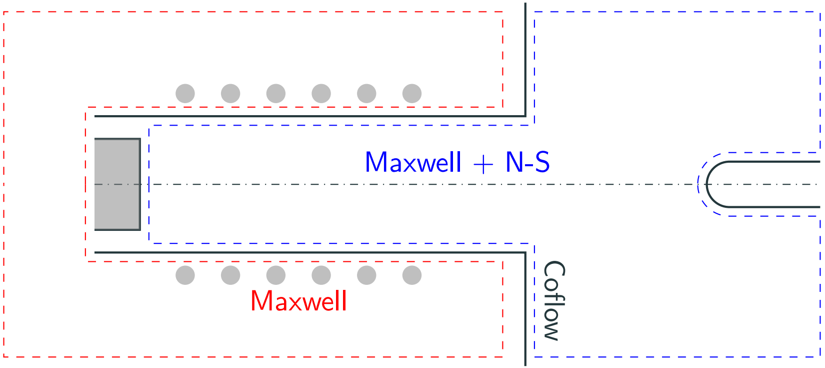
No-slip wall isothermal $T = T_{wall}$ and $\mathbf{u} = \mathbf{0}$.

Computational domain



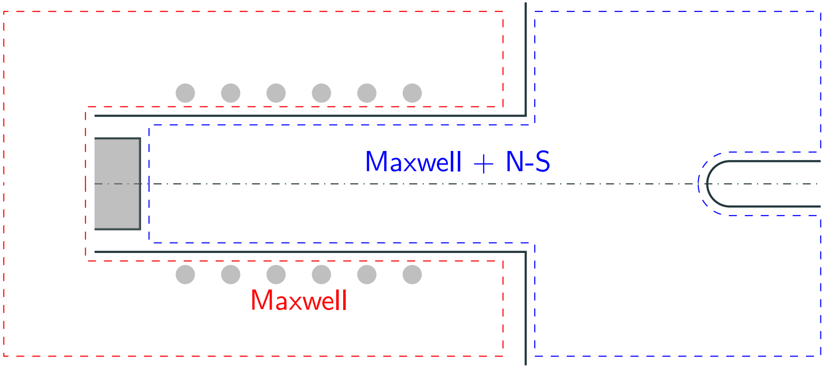
Inflow $u_z = U_{in}$, $u_r = 0$ and $T = T_{inlet}$.

Computational domain



Stabilizing co-flow $T = T_{wall}$, $\mathbf{u} = u_{coflow}\mathbf{e}_z$

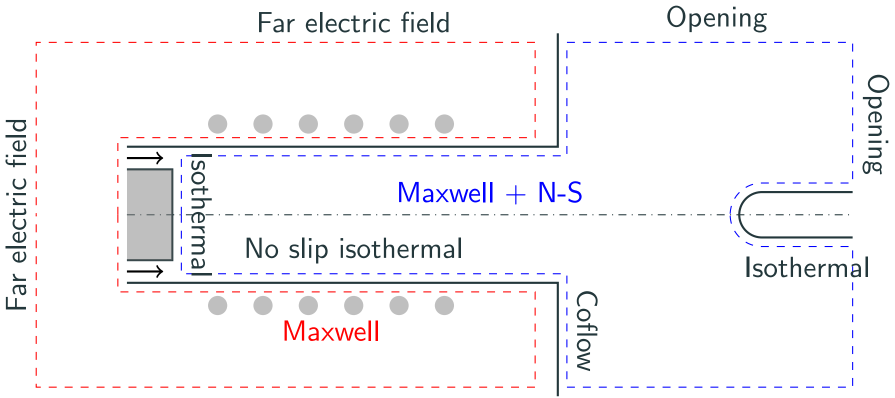
Computational domain



ICP and Maxwell interface

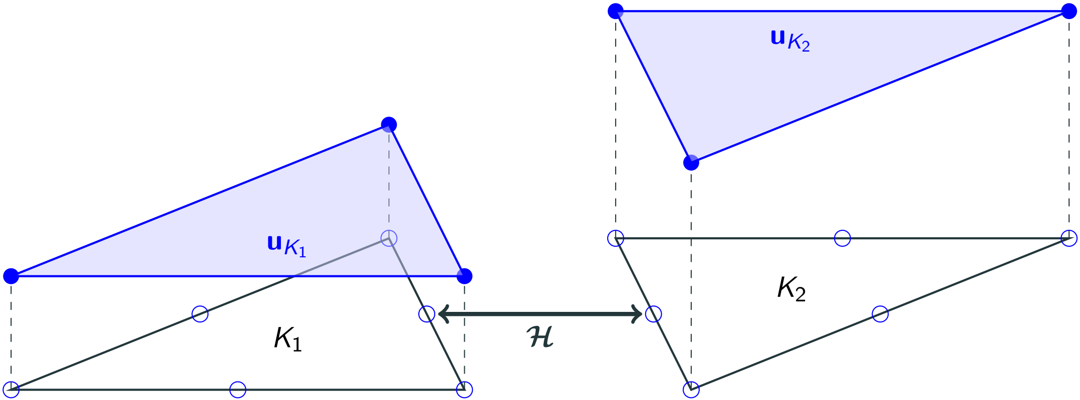
$$
\begin{aligned}
\nabla E_p^{Re,NS} \cdot \mathbf{n}^{NS} &= \nabla E_p^{Re,MAX} \cdot \mathbf{n}^{MAX} \\
\nabla E_p^{Im,NS} \cdot \mathbf{n}^{NS} &= \nabla E_p^{Im,MAX} \cdot \mathbf{n}^{MAX}
\end{aligned}
$$

Computational domain



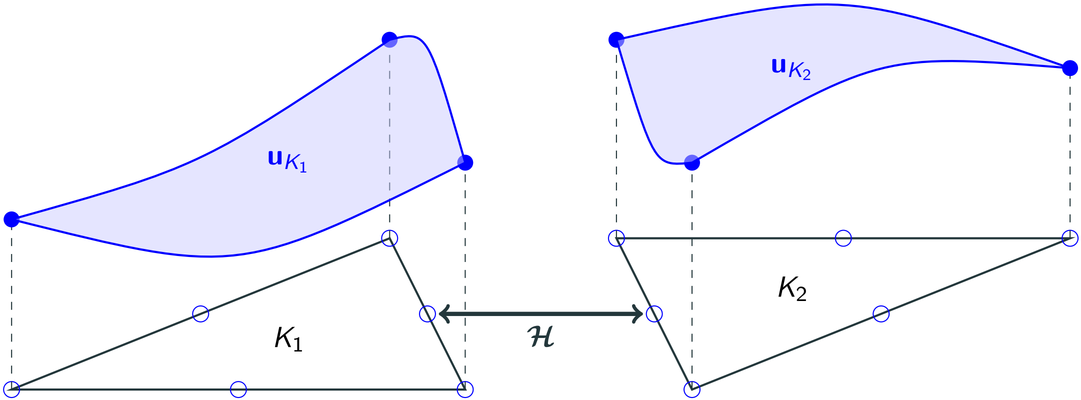
Hybridized discontinuous Galerkin method

Finite volume and (hybridized) discontinuous Galerkin



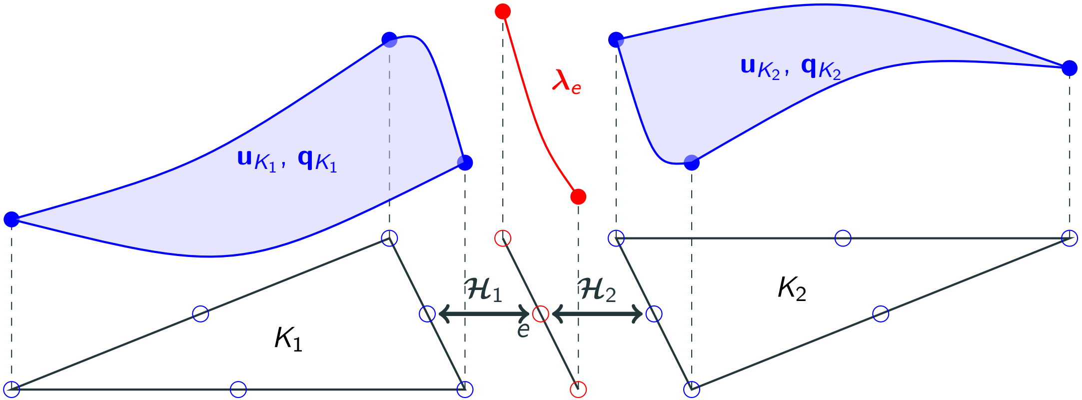
Finite volumes

Finite volume and (hybridized) discontinuous Galerkin



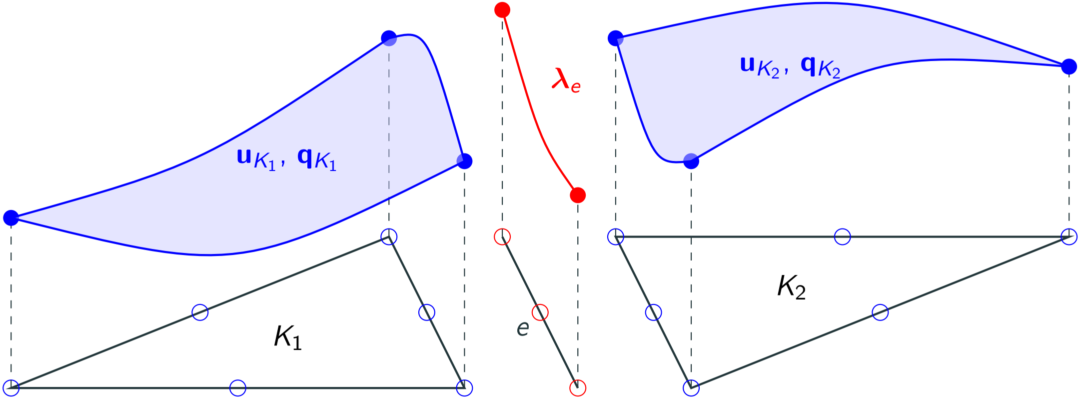
Discontinuous Galerkin

Finite volume and (hybridized) discontinuous Galerkin



Hybridized discontinuous Galerkin

Finite volume and (hybridized) discontinuous Galerkin



Why HDG over DG? As will be seen later, HDG allows for static condensation, effectively reducing the number of DOFs when using a Newton solver.

Conservative form of the equations

$$
\begin{aligned}
&\partial_t \rho &&+ \nabla \cdot (\rho \mathbf{v}) && &&= 0\\
&\partial_t (\rho \mathbf{v}) &&+ \nabla \cdot (\rho \mathbf{v}\mathbf{v} + p\mathbb{I}) &&- \nabla \cdot \boldsymbol{\tau} &&= \mathbf{F}_L\\
&\partial_t (\rho E) &&+ \nabla \cdot (\rho E \mathbf{v} + p\mathbf{v}) &&- \nabla \cdot (\boldsymbol{\tau}\mathbf{v} + \mathbf{q}) &&= P_J\\
& && &&- \nabla \cdot (\nabla E_P) &&= -\frac{E_P}{r^2} - i2\pi f \mu_0 \sigma_e (E_C + E_P)
\end{aligned}
$$

Conservative form of the equations

$$
\begin{aligned}
&\partial_t \rho && + \nabla \cdot (\rho \mathbf{v}) && && = 0 \\
&\partial_t (\rho \mathbf{v}) && + \nabla \cdot (\rho \mathbf{v}\mathbf{v} + p\mathbb{I}) && - \nabla \cdot \boldsymbol{\tau} && = \mathbf{F}_L \\
&\partial_t (\rho E) && + \nabla \cdot (\rho E \mathbf{v} + p\mathbf{v}) && - \nabla \cdot (\boldsymbol{\tau}\mathbf{v} + \mathbf{q}) && = P_J \\
& && && - \nabla \cdot (\nabla E_P) && = -\frac{E_P}{r^2} - i2\pi f \mu_0 \sigma_e (E_C + E_P)
\end{aligned}
$$

$$
\partial_t \mathbf{w}(\mathbf{u}) \quad + \nabla \cdot \mathbf{F}_c(\mathbf{u}) \quad - \nabla \cdot \mathbf{F}_v(\mathbf{u}, \nabla \mathbf{u}) \quad = \mathbf{S}(\mathbf{u}, \nabla \mathbf{u})
$$

Conservative form of the equations

$$
\begin{aligned}
&\partial_t \rho && + \nabla \cdot (\rho \mathbf{v}) && && = 0 \\
&\partial_t (\rho \mathbf{v}) && + \nabla \cdot (\rho \mathbf{v}\mathbf{v} + p\mathbb{I}) && - \nabla \cdot \boldsymbol{\tau} && = \mathbf{F}_L \\
&\partial_t (\rho E) && + \nabla \cdot (\rho E \mathbf{v} + p\mathbf{v}) && - \nabla \cdot (\boldsymbol{\tau}\mathbf{v} + \mathbf{q}) && = P_J \\
& && && - \nabla \cdot (\nabla E_P) && = -\frac{E_P}{r^2} - i2\pi f \mu_0 \sigma_e (E_C + E_P)
\end{aligned}
$$

$$
\boxed{\partial_t \mathbf{w}(\mathbf{u}) + \nabla \cdot \mathbf{F}_c(\mathbf{u}) - \nabla \cdot \mathbf{F}_v(\mathbf{u}, \nabla \mathbf{u}) = \mathbf{S}(\mathbf{u}, \nabla \mathbf{u})}
$$

Model problem

$$
\begin{aligned}
&\partial_t \mathbf{w}(\mathbf{u}) + \nabla \cdot (\mathbf{F}_c(\mathbf{u}) - \mathbf{F}_v(\mathbf{u}, \nabla \mathbf{u})) = \mathbf{S}(\mathbf{u}, \nabla \mathbf{u}) && \text{on } \Omega \\
&\mathbf{u} = \mathbf{u}_{bc} && \text{on } \partial\Omega_D \\
&\mathbf{F}_v \cdot \mathbf{n} = \mathbf{F}_{v,n,bc} && \text{on } \partial\Omega_N \\
&\mathbf{u}(t = 0) = \mathbf{U} && \text{on } \Omega
\end{aligned}
$$

$\mathbf{F}_{c,v}$: convective and diffusive fluxes.

$\mathbf{S}$: source terms.

$\mathbf{w}$: conservative variables.

$\mathbf{u}$, $\mathbf{u}_{bc}$: solution and boundary condition.

Ω : domain of the problem.

$\partial\Omega_{D,N}$: domain boundary for Dirichlet and Neumann BC.

$\mathbf{U}$: initial data.

Model problem

$$\begin{aligned}
&\partial_t \mathbf{w}(\mathbf{u}) + \nabla \cdot (\mathbf{F}_c(\mathbf{u}) - \mathbf{F}_v(\mathbf{u}, \nabla \mathbf{u})) = \mathbf{S}(\mathbf{u}, \nabla \mathbf{u}) && \text{on } \Omega \\
&\mathbf{u} = \mathbf{u}_{bc} && \text{on } \partial\Omega_D \\
&\mathbf{F}_v \cdot \mathbf{n} = \mathbf{F}_{v,n,bc} && \text{on } \partial\Omega_N \\
&\mathbf{u}(t = 0) = \mathbf{U} && \text{on } \Omega
\end{aligned}$$

Before going further, some notation:

$$\begin{aligned}
(a, b)_K &= \int_K ab \, dV, \\
\langle a, b \rangle_{\partial K} &= \int_{\partial K} ab \, dS.
\end{aligned}$$

From weak form to HDG

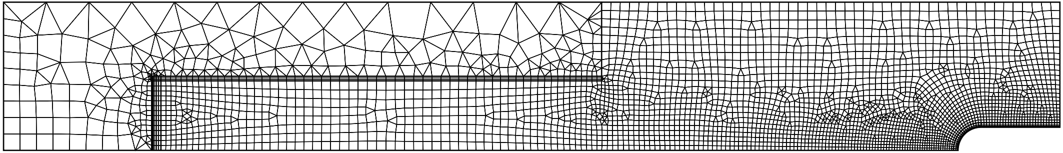
$$(\partial_t \mathbf{w} - \mathbf{S}, w)_\Omega - (\mathbf{F}_c - \mathbf{F}_v, \nabla w)_\Omega + \langle (\mathbf{F}_c - \mathbf{F}_v) \cdot \mathbf{n}, w \rangle_{\partial\Omega} = 0, \ \forall w \in L_2(\Omega)$$

Multiplying by a function $w \in L_2(\Omega)$ and integrating over the domain gives the **weak form**.

From weak form to HDG

$$(\partial_t \mathbf{w} - \mathbf{S}, w)_{\mathcal{T}} - (\mathbf{F}_c - \mathbf{F}_v, \nabla w)_{\mathcal{T}} + \sum_{K \in \mathcal{T}} \langle (\mathbf{F}_c - \mathbf{F}_v) \cdot \mathbf{n}, w \rangle_{\partial K} = 0, \ \forall w \in L_2(\Omega)$$

The domain Ω is split in a collection of elements $K \in \mathcal{T}$.



From weak form to HDG

$$(\partial_t \mathbf{w} - \mathbf{S}, w)_{\mathcal{T}} - (\mathbf{F}_c - \mathbf{F}_v, \nabla w)_{\mathcal{T}} + \sum_{K\in\mathcal{T}} \langle (\mathbf{F}_c - \mathbf{F}_v) \cdot \mathbf{n}, w \rangle_{\partial K} = 0, \; \forall w \in W_h$$

We restrict w to a **finite dimensional** subset of $L_2(\Omega)$.

$$W_h = \left\{ w \in L^2(\Omega) : w|_K \in \mathcal{P}^p(K), \forall K \in \mathcal{T} \right\} \subset L_2(\Omega)$$

From weak form to HDG

$$(\partial_t \mathbf{w} - \mathbf{S}, w)_{\mathcal{T}} - (\mathbf{F}_c - \mathbf{F}_v, \nabla w)_{\mathcal{T}} + \sum_{K \in \mathcal{T}} \langle \boldsymbol{\mathcal{H}}(\mathbf{u}, \nabla \mathbf{u}), w \rangle_{\partial K} = 0, \ \forall w \in W_h$$

For stabilizing the method, the convective and diffusive fluxes on the interface are approximated

$$(\mathbf{F}_c - \mathbf{F}_v) \cdot \mathbf{n}_K \simeq \boldsymbol{\mathcal{H}}$$

From weak form to HDG

$$(\partial_t \mathbf{w}_h - \mathbf{S}_h, w)_{\mathcal{T}} - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla w)_{\mathcal{T}} + \sum_{K \in \mathcal{T}} \langle \boldsymbol{\mathcal{H}}(\mathbf{u}, \nabla \mathbf{u}), w \rangle_{\partial K} = 0, \ \forall w \in W_h$$

The solution is also approximated by

$$\mathbf{u} \simeq \mathbf{u}_h \in W_h$$

From weak form to HDG

$$
(\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla \varphi_{K,i})_K + \langle \boldsymbol{\mathcal{H}}(\mathbf{u}_h, \nabla \mathbf{u}_h), \varphi_{K,i} \rangle_{\partial K} = 0,
$$
$$
\forall \varphi_{K,i} \in W_h
$$

If the $\varphi_{K,i}$ is a local basis of W_h on K, $i \in \{1, 2, ..., p\}$,

$$
\mathbf{u} \simeq \mathbf{u}_h = \sum_{K \in \mathcal{T}} \sum_{i=1}^{p} \mathbf{u}_{K,i} \varphi_{K,i}
$$

From weak form to HDG

$$(\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla\varphi_{K,i})_K + \langle \boldsymbol{\mathcal{H}}(\mathbf{u}_h, \nabla\mathbf{u}_h), \varphi_{K,i}\rangle_{\partial K} = 0,$$
$$\forall \varphi_{K,i} \in W_h$$

- $\varphi_{K,i}$ are discontinuous across elements.
- If $\varphi_{K,i} = 1$ everywhere $\Rightarrow$ finite volume.

From weak form to HDG

$$
\begin{aligned}
&(\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla \varphi_{K,i})_K + \langle \boldsymbol{\mathcal{H}}(\mathbf{u}_h, \mathbf{q}_h), \varphi_{K,i} \rangle_{\partial K} = 0,\\
&(\mathbf{q}_h, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_h, \nabla \boldsymbol{\tau}_{K,i})_K + \langle \mathbf{u}_h, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_{bc}} = 0,\\
&\forall \varphi_{K,i} \in W_h, \ \forall \boldsymbol{\tau}_{K,i} \in V_h
\end{aligned}
$$

With the solution gradient

$$\mathbf{q} = \nabla \mathbf{u} \simeq \mathbf{q}_h$$

with the subset

$$V_h = \left\{ \mathbf{v} \in L^2(\Omega) : \mathbf{v}|_K \in \left(\mathcal{P}^p(K)\right)^D, \forall K \in \mathcal{T} \right\}$$

and local basis $\boldsymbol{\tau}_{K,i}$.

From weak form to HDG

$$
\begin{aligned}
&(\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla \varphi_{K,i})_K + \langle \boldsymbol{\mathcal{H}}(\boldsymbol{\lambda}_h, \mathbf{u}_h, \mathbf{q}_h), \varphi_{K,i} \rangle_{\partial K} = 0,\\
&(\mathbf{q}_h, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_h, \nabla \boldsymbol{\tau}_{K,i})_K + \langle \boldsymbol{\lambda}_h, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_{bc}} = 0\\
&\forall \varphi_{K,i} \in W_h,\ \forall \boldsymbol{\tau}_{K,i} \in V_h
\end{aligned}
$$

$$\Gamma = \{e : e = K_i \cap K_j; \forall K_i, K_j \in \mathcal{T},\ K_i \neq K_j\}.$$

$$M_h = \left\{\mu \in L^2(\Gamma) : \mu|_e \in \mathcal{P}^p(e), \forall e \in \Gamma\right\}$$

$$\boldsymbol{\lambda}_h = \sum_{e \in \Gamma} \sum_{i=1}^{p} \boldsymbol{\lambda}_{e,i} \mu_{e,i}$$

From weak form to HDG

$$
\begin{aligned}
&(\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla \varphi_{K,i})_K + \langle \boldsymbol{\mathcal{H}}(\boldsymbol{\lambda}_h, \mathbf{u}_h, \mathbf{q}_h), \varphi_{K,i} \rangle_{\partial K} = 0,\\
&(\mathbf{q}_h, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_h, \nabla \boldsymbol{\tau}_{K,i})_K + \langle \boldsymbol{\lambda}_h, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_{bc}} = 0,\\
&\langle [[\boldsymbol{\mathcal{H}}]], \mu_{e,i} \rangle_e = 0,\\
&\forall \mu_{e,i} \in M_h,\ \forall \varphi_{K,i} \in W_h,\ \forall \boldsymbol{\tau}_{K,i} \in V_h
\end{aligned}
$$

Equation for $\boldsymbol{\lambda}$: conservativity of the normal numerical flux across the interfaces.

The jump operator is given by

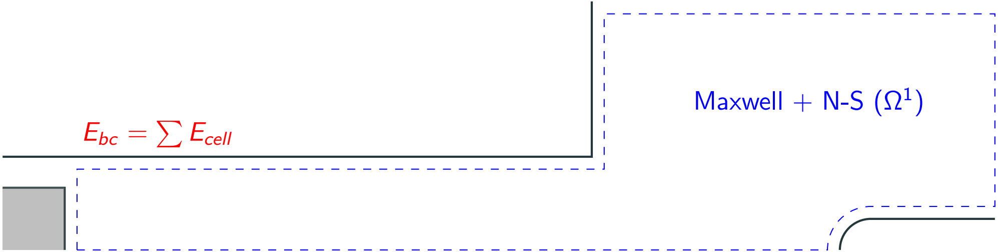
$$[[\mathbf{u}]] = \mathbf{u}_+ \cdot \mathbf{n}_+ + \mathbf{u}_- \cdot \mathbf{n}_-$$

From weak form to HDG

$$
\begin{aligned}
&(\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla \varphi_{K,i})_K + \langle \boldsymbol{\mathcal{H}}(\boldsymbol{\lambda}_h, \mathbf{u}_h, \mathbf{q}_h), \varphi_{K,i} \rangle_{\partial K} = 0, \\
&(\mathbf{q}_h, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_h, \nabla \boldsymbol{\tau}_{K,i})_K + \langle \boldsymbol{\lambda}_h, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_{bc}} = 0, \\
&\langle [[\boldsymbol{\mathcal{H}}]], \mu_{e,i} \rangle_e = 0, \\
&\forall \mu_{e,i} \in M_h,\ \forall \varphi_{K,i} \in W_h,\ \forall \boldsymbol{\tau}_{K,i} \in V_h
\end{aligned}
$$

ICP as a multi-domain problem

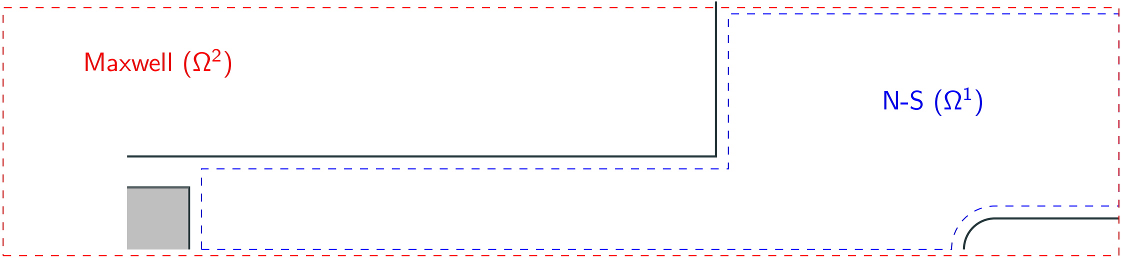
Single domain + integral boundary value



Compact but fills the jacobian matrix, making it incompatible with HDG.

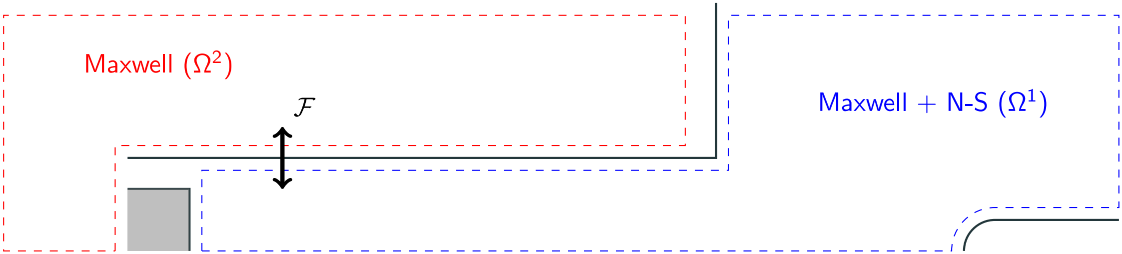
ICP as a multi-domain problem

Two overlapping domain



Most widely used. Solve N-S while Maxwell frozen, and *vice versa*. Slow convergence.

Multi-domain



Solves the system in a fully coupled manner, with two domains. The method we employ here.

Modification of the model problem

$$
\begin{aligned}
&\partial_t \mathbf{w}^l(\mathbf{u}) + \nabla \cdot \Big(\mathbf{F}_c^l(\mathbf{u}) - \mathbf{F}_v^l(\mathbf{u}, \nabla \mathbf{u})\Big) = \mathbf{S}^l(\mathbf{u}, \nabla \mathbf{u}), && \text{on } \Omega^l \\
&\mathbf{u}^l = \mathbf{u}_{bc}^l, && \mathbf{x} \in \partial\Omega_d^l \\
&\mathbf{F}_v^l \cdot \mathbf{n} = \mathbf{F}_{v,n,bc}^l, && \text{on } \partial\Omega_n^l \\
&\mathcal{F}^{l,l'}(\mathbf{u}^l, \nabla \mathbf{u}^l, \mathbf{u}^{l'}, \nabla \mathbf{u}^{l'}) = 0, && \text{on } \gamma^{l,l'},\ l \neq l' \\
&\mathbf{u}^l(t=0) = \mathbf{U}^l, && \text{on } \Omega^l
\end{aligned}
$$

Modification of the model problem

$$
\begin{aligned}
&\partial_t \mathbf{w}^l(\mathbf{u}) + \nabla \cdot \Big(\mathbf{F}_c^l(\mathbf{u}) - \mathbf{F}_v^l(\mathbf{u}, \nabla \mathbf{u})\Big) = \mathbf{S}^l(\mathbf{u}, \nabla \mathbf{u}), && \text{on } \Omega^l \\
&\mathbf{u}^l = \mathbf{u}_{bc}^l, && \mathbf{x} \in \partial\Omega_d^l \\
&\mathbf{F}_v^l \cdot \mathbf{n} = \mathbf{F}_{v,n,bc}^l, && \text{on } \partial\Omega_n^l \\
&{\color{red}\mathcal{F}^{l,l'}(\mathbf{u}^l, \nabla \mathbf{u}^l, \mathbf{u}^{l'}, \nabla \mathbf{u}^{l'}) = 0}, && \text{on } \gamma^{l,l'},\ l \neq l' \\
&\mathbf{u}^l(t=0) = \mathbf{U}^l, && \text{on } \Omega^l
\end{aligned}
$$

Modification of the model problem

$$\begin{aligned}
&\partial_t \mathbf{w}^l(\mathbf{u}) + \nabla \cdot \Big(\mathbf{F}_c^l(\mathbf{u}) - \mathbf{F}_v^l(\mathbf{u}, \nabla \mathbf{u})\Big) = \mathbf{S}^l(\mathbf{u}, \nabla \mathbf{u}), && \text{on } \Omega^l \\
&\mathbf{u}^l = \mathbf{u}_{bc}^l, && \mathbf{x} \in \partial\Omega_d^l \\
&\mathbf{F}_v^l \cdot \mathbf{n} = \mathbf{F}_{v,n,bc}^l, && \text{on } \partial\Omega_n^l \\
&\mathcal{F}^{l,l'}(\mathbf{u}^l, \nabla \mathbf{u}^l, \mathbf{u}^{l'}, \nabla \mathbf{u}^{l'}) = 0, && \text{on } \gamma^{l,l'},\ l \neq l' \\
&\mathbf{u}^l(t=0) = \mathbf{U}^l, && \text{on } \Omega^l
\end{aligned}$$

For ICP,

$$\mathcal{F} = \begin{pmatrix} \nabla E_p^{Re,NS} \cdot \mathbf{n}^{NS} - \nabla E_p^{Re,MAX} \cdot \mathbf{n}^{MAX} \\ \nabla E_p^{Im,NS} \cdot \mathbf{n}^{NS} - \nabla E_p^{Im,MAX} \cdot \mathbf{n}^{MAX} \end{pmatrix}$$

Modification of the discrete formulation

$$\left(\partial_t \mathbf{w}_h - \mathbf{S}_h^K, \varphi_{K,i}\right)_K - \left(\mathbf{F}_{h,c}^K - \mathbf{F}_{h,v}^K, \nabla\varphi_{K,i}\right)_K + \left\langle \boldsymbol{\mathcal{H}}^K, \varphi_{K,i} \right\rangle_{\partial K} = 0$$

$$\left(\mathbf{q}_K, \boldsymbol{\tau}_{K,i}\right)_K - \left(\mathbf{u}_K, \nabla\boldsymbol{\tau}_{K',i}\right)_K + \langle \boldsymbol{\lambda}_{\partial K}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n}\rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n}\rangle_{\partial K_{bc}} = 0$$

$$\langle [[\boldsymbol{\mathcal{H}}]], \mu_{e,i}\rangle_{e \setminus \bar{\Gamma}} + \langle \tilde{\mathcal{F}}^e, \mu_{e,i}\rangle_{e \cap \bar{\Gamma}} = 0$$

with $\bar{\Gamma}$ the set of interfaces between subdomains.

Numerical fluxes : convection

$$\boldsymbol{\mathcal{H}} = \boldsymbol{\mathcal{H}}_c - \boldsymbol{\mathcal{H}}_v$$

$$\boldsymbol{\mathcal{H}}_c = R\boldsymbol{\mathcal{H}}_{c,n}(\mathbf{u}, \boldsymbol{\lambda}) = R(\dot{m} + \dot{m}_p)\Psi_\lambda - R|\dot{m}|(\Psi_\lambda - \Psi_U) + \mathbf{P}$$

with

$$\boldsymbol{\Psi} = \begin{pmatrix} 1 & v^\perp & v^\parallel & v_\theta & e + \frac{1}{2}||\mathbf{v}||^2 + \frac{p}{\rho} \end{pmatrix}^T$$

$$\mathbf{P} = \begin{pmatrix} 0 & p_\lambda & 0 & 0 \end{pmatrix}^T$$

Numerical fluxes : diffusion

$$\boxed{\mathcal{H} = \mathcal{H}_c - \mathcal{H}_v}$$

$$\mathcal{H}_v(\mathbf{u}, \lambda, \mathbf{q}) = \mathbf{F}_v(\boldsymbol{\lambda}, \mathbf{q}) \cdot \mathbf{n} + \gamma(h)(\boldsymbol{\lambda} - \mathbf{u})$$

with

$$\gamma = C \max_{f \in K} \left(\frac{1}{2\mathcal{V}} \sum_{f \in K} \mathcal{A} \right)$$

C depends on the diffusive transport coefficients.

Solution strategy

The HDG system is solved at steady-state such that

$$
\begin{aligned}
&\left(\cancel{\partial_t \mathbf{w}_h} - \mathbf{S}_h^K, \varphi_{K,i}\right)_K - \left(\mathbf{F}_{h,c}^K - \mathbf{F}_{h,v}^K, \nabla\varphi_{K,i}\right)_K + \left\langle \boldsymbol{\mathcal{H}}^K, \varphi_{K,i}\right\rangle_{\partial K} = 0\\
&(\mathbf{q}_K, \boldsymbol{\tau}_{K,i})_K - \left(\mathbf{u}_K, \nabla\boldsymbol{\tau}_{K',i}\right)_K + \langle \boldsymbol{\lambda}_{\partial K}, \boldsymbol{\tau}_{K,i}\cdot\mathbf{n}\rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i}\cdot\mathbf{n}\rangle_{\partial K_{bc}} = 0\\
&\langle [[\boldsymbol{\mathcal{H}}]], \mu_{e,i}\rangle_{e\setminus\bar{\Gamma}} + \langle \tilde{\mathcal{F}}^e, \mu_{e,i}\rangle_{e\cap\bar{\Gamma}} = 0
\end{aligned}
$$

If the system is written as $\mathcal{N} = 0$, then the Newton method solves

$$\mathcal{N}'(\mathbf{x}_k)\delta\mathbf{x}_k = -\mathcal{N}(\mathbf{x}_k)$$

with $\delta\mathbf{x}_k = \mathbf{x}_{k+1} - \mathbf{x}_k$.

Solution strategy

The HDG system is solved at steady-state such that

$$\left(\frac{\delta \mathbf{w}_h^K}{\tau_{K,k}} - \mathbf{S}_h^K, \varphi_{K,i}\right)_K - \left(\mathbf{F}_{h,c}^K - \mathbf{F}_{h,v}^K, \nabla\varphi_{K,i}\right)_K + \left\langle \boldsymbol{\mathcal{H}}^K, \varphi_{K,i}\right\rangle_{\partial K} = 0$$

$$(\mathbf{q}_K, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_K, \nabla \boldsymbol{\tau}_{K',i})_K + \langle \boldsymbol{\lambda}_{\partial K}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n}\rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n}\rangle_{\partial K_{bc}} = 0$$

$$\langle [[\boldsymbol{\mathcal{H}}]], \mu_{e,i}\rangle_{e\setminus\bar{\Gamma}} + \langle \tilde{\mathcal{F}}^e, \mu_{e,i}\rangle_{e\cap\bar{\Gamma}} = 0$$

If the system is written as $\tilde{\mathcal{N}} = 0$, then the Newton method solves

$$\tilde{\mathcal{N}}'(\mathbf{x}_k)\delta\mathbf{x} = -\tilde{\mathcal{N}}(\mathbf{x}_k)$$

with $\delta\mathbf{x} = \mathbf{x}_{k+1} - \mathbf{x}_k$.

In order to stabilize the solver, a fictive temporal term is added.

Damping term

$$\frac{\delta \mathbf{w}_h}{\tau_K} = \frac{\mathbf{w}(\mathbf{u}_{h,k+1}) - \mathbf{w}(\mathbf{u}_{h,k})}{\tau_{K,k}}$$

with

$$\tau_{K,k} = CFL_k \frac{h_K}{v_K} = CFL_k \frac{V_K}{\int_{\partial K} \lambda_{max} dS}$$

and the following CFL evolution law

$$CFL_k = \min\left(CFL_0 \left(\frac{||\tilde{\mathcal{N}}||_2^0}{||\tilde{\mathcal{N}}||_2^k} \right)^{\alpha}, CFL_{max} \right)$$

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We only use convective eigenvalues here. The diffusive could be use to enhance the convergence.

Matrix form of the system

$$\boxed{\tilde{\mathcal{N}}'(\mathbf{x}_k)\delta\mathbf{x}_k = -\tilde{\mathcal{N}}(\mathbf{x}_k)}$$

Matrix form of the system

$$\boxed{\tilde{\mathcal{N}}'(\mathbf{x}_k)\delta\mathbf{x}_k = -\tilde{\mathcal{N}}(\mathbf{x}_k)}$$

$$\underbrace{\begin{pmatrix} A & B & R \\ C & D & S \\ L & M & N \end{pmatrix}}_{\tilde{\mathcal{N}}'} \overbrace{\begin{pmatrix} \delta Q \\ \delta U \\ \delta \Lambda \end{pmatrix}}^{\delta\mathbf{x}} = \underbrace{\begin{pmatrix} F \\ G \\ H \end{pmatrix}}_{-\tilde{\mathcal{N}}},$$

Matrix form of the system

$$\boxed{\tilde{\mathcal{N}}'(\mathbf{x}_k)\delta\mathbf{x}_k = -\tilde{\mathcal{N}}(\mathbf{x}_k)}$$

The system can be split into two parts

$$\underbrace{\begin{pmatrix} A & B \\ C & D \end{pmatrix}}_{\text{Block diagonal}} \begin{pmatrix} \delta Q \\ \delta U \end{pmatrix} = \begin{pmatrix} F \\ G \end{pmatrix} - \begin{pmatrix} R \\ S \end{pmatrix} \delta\Lambda$$

and

$$\begin{pmatrix} L & M \end{pmatrix} \begin{pmatrix} \delta Q \\ \delta U \end{pmatrix} + N\delta\Lambda = H$$

Matrix form of the system

$$\tilde{\mathcal{N}}'(\mathbf{x}_k)\delta\mathbf{x}_k = -\tilde{\mathcal{N}}(\mathbf{x}_k)$$

By eliminating δU and δQ, one gets the **global** system

$$\left(N - \begin{pmatrix} L & M \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} R \\ S \end{pmatrix}\right) \delta\Lambda = H - \begin{pmatrix} L & M \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} F \\ G \end{pmatrix}$$

Matrix form of the system

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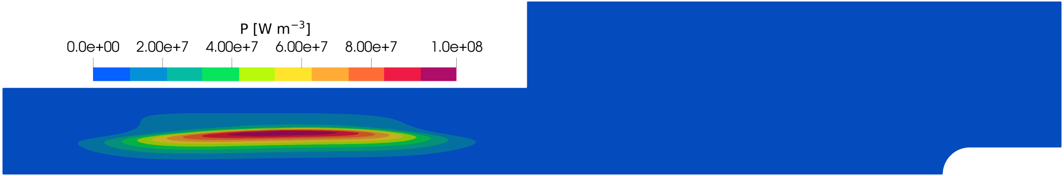
System solved for $\boldsymbol{\lambda}$, then $\mathbf{u}$ and $\mathbf{q}$ are retrieved by small local matrices inversion.

$$\begin{pmatrix} \delta Q \\ \delta U \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} F \\ G \end{pmatrix} - \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} R \\ S \end{pmatrix} \delta\Lambda$$

The problem of power quenching

If the power dissipated is not maintained, the torch quenches!



$$P_{target} = \int_{NS} P_J dV$$

The problem of power quenching

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$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v}\mathbf{v}) + \nabla p - \nabla \cdot \boldsymbol{\tau} = \mathbf{F}^L$$

$$\partial_t \left(\rho e + \frac{1}{2}\rho ||\mathbf{v}||^2 \right) + \nabla \cdot \left(\rho e \mathbf{v} + \frac{1}{2}\rho ||\mathbf{v}||^2 \mathbf{v} + p\mathbf{v} - \boldsymbol{\tau} \cdot \mathbf{v} - \mathbf{q} \right) = P_J$$

$$\Delta E_P = \frac{E_P}{r^2} + i 2\pi f \mu_0 \sigma_e (E_C + E_P)$$

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$$\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v}\mathbf{v}) + \nabla p - \nabla \cdot \boldsymbol{\tau} = \gamma \left(\mathbf{F}^L\right)'$$

$$\partial_t \left(\rho e + \frac{1}{2}\rho||\mathbf{v}||^2\right) + \nabla \cdot \left(\rho e \mathbf{v} + \frac{1}{2}\rho||\mathbf{v}||^2 \mathbf{v} + p\mathbf{v} - \boldsymbol{\tau} \cdot \mathbf{v} - \mathbf{q}\right) = \gamma P_J'$$

$$\Delta E_P' = \frac{E_P'}{r^2} + i2\pi f \mu_0 \sigma_e \left(E_C' + E_P'\right)$$

with

$$E_P = \sqrt{\gamma} E_P' \quad E_C = \sqrt{\gamma} E_C'$$

Power ratio

γ is computed at every Newton iteration by equalizing

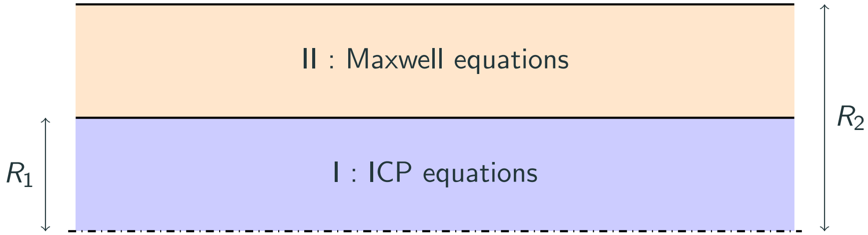
$$\gamma = \frac{P_{target}}{\int_{NS} P'_J dV}$$

with P_{target} the desired power dissipated in the facility.

The power was found to remain constant and γ was converging towards a value using this method.

Results

Manufactured solution

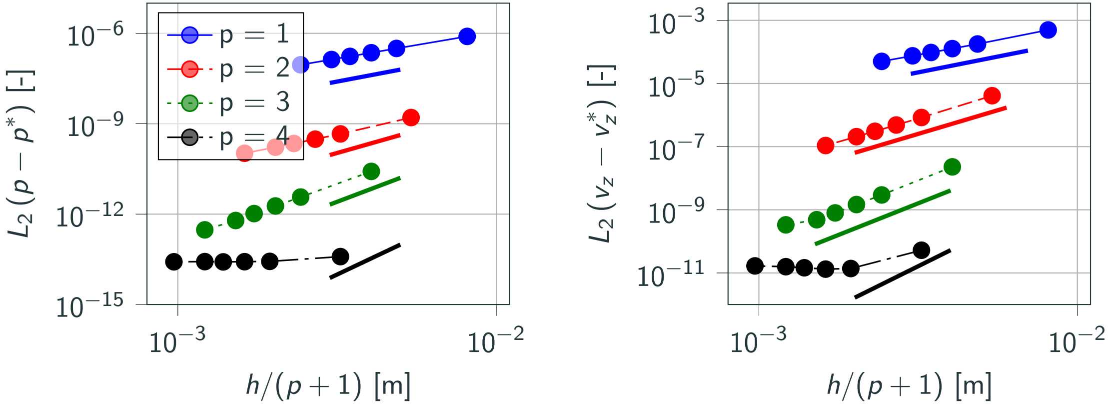


$$p = \Delta p_0 f(z,r) + p_0 \qquad T = T_{min} + (T_{max} - T_{min})\, f(z,r)$$
$$v_z = u_0 f(z,r) \qquad E_p = E_0 f(z,r)(1-i)$$
$$v_r = u_0 f(z,r) \qquad v_\theta = u_0 f(z,r)$$

with

$$f(z,r) = \left(\frac{rz}{RL}\right)^2 \exp\left(-\frac{z}{L} - \frac{r}{R_1}\right)$$

Convergence study : in the torch



Convergence order of $p+1$ is retrieved!

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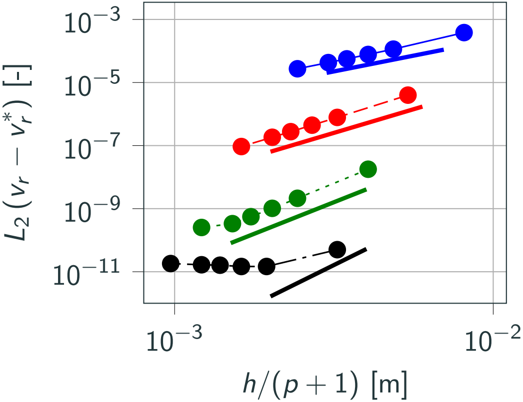

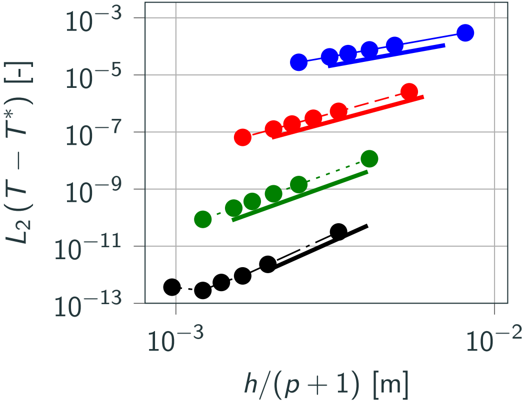

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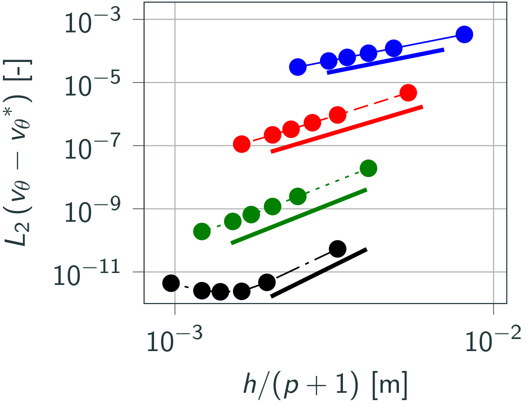

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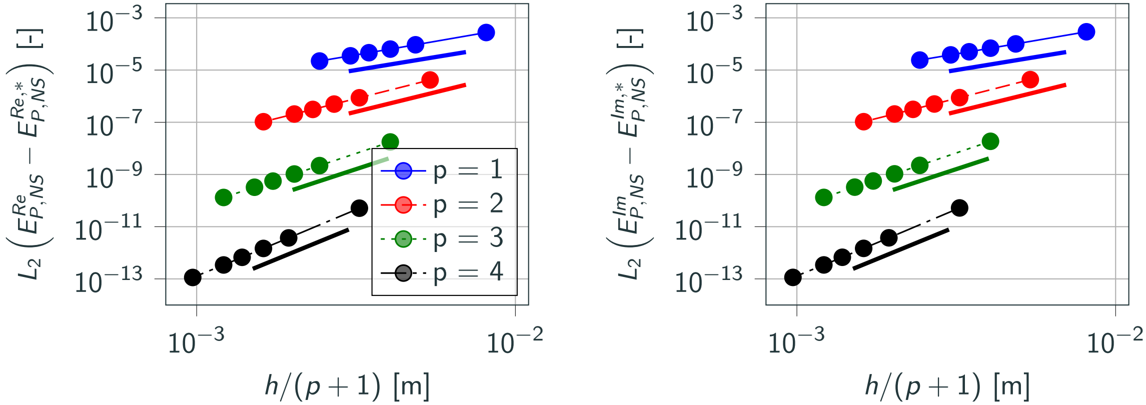

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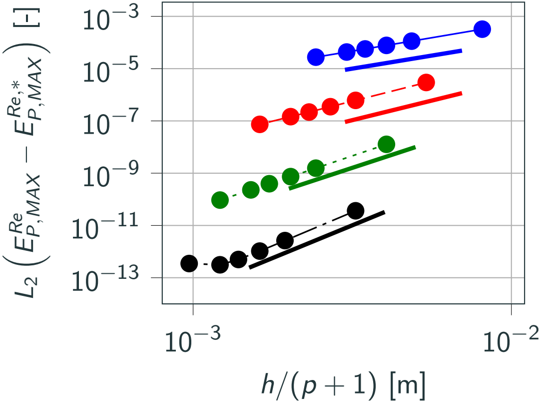
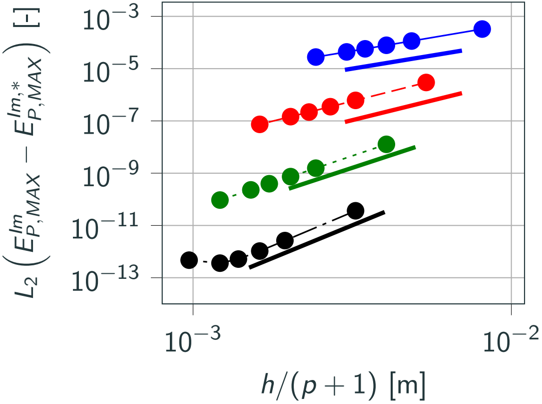
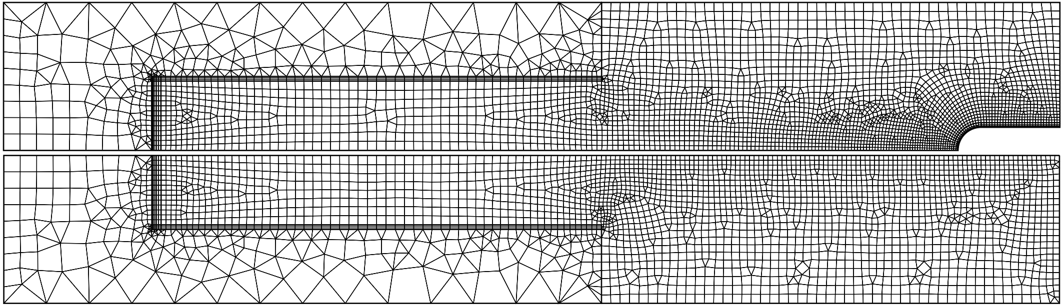




Convergence order of $p + 1$ is retrieved!

Mesh and order dependence study



This mesh at $p = 4$ with 4013 elements for the probe case and 3055 for the freestream case is sufficient for capturing the solution accurately.

The mesh size close to the probe is 10 µm thick, while it is of 1 µm for FV!

The question of the coflow

A coflow is introduced in the chamber for obtaining a steady state, does it have an important impact?

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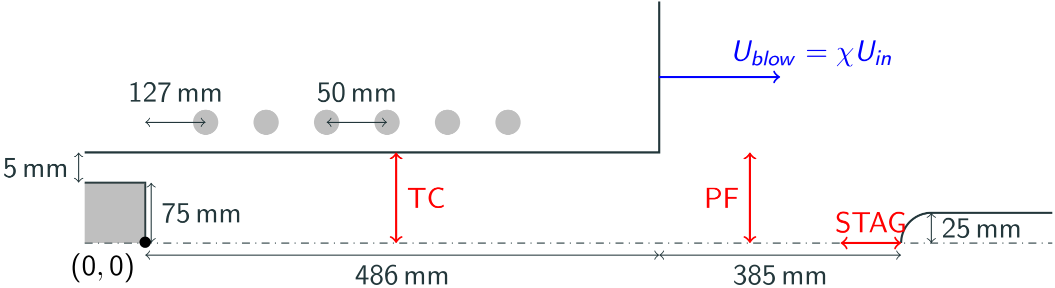


The question of the coflow

A coflow is introduced in the chamber for obtaining a steady state, does it have an important impact?

- $P = 100\,\mathrm{kW}$
- $p_0 = 10\,000\,\mathrm{Pa}$
- $f = 0.37\,\mathrm{MHz}$
- $Q = 16\,\mathrm{g\,s^{-1}}$
- $S = 20^\circ$
- air mixture with 11 species

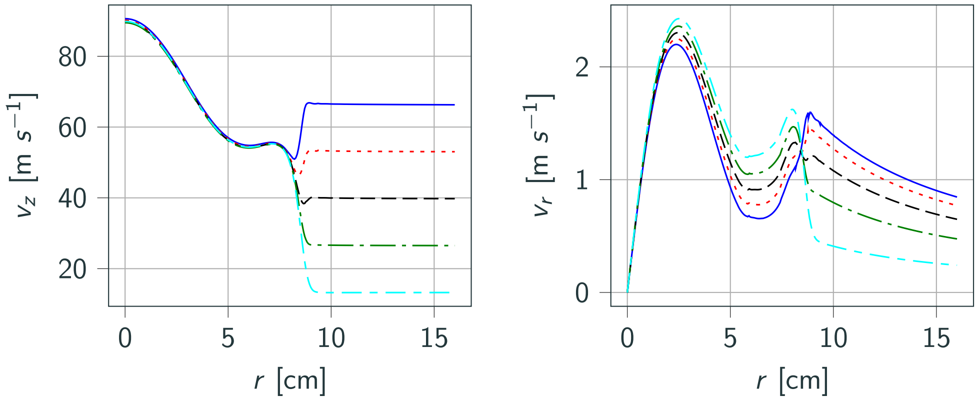
The coflow χ is defined as

$$\chi = \frac{U_{blow}}{U_{in}}$$

and tested from 0.2 to 1.0.

The question of the coflow

A coflow is introduced in the chamber for obtaining a steady state, does it have an important impact?



The question of the coflow

A coflow is introduced in the chamber for obtaining a steady state, does it have an important impact? Short answer is no, BUT...

... we were not able to obtain any steady state without coflow.

A question to ask is to see if steady state is possible in ICP?

Left for future work...

Comparison with the COOLFluiD solver

We compared the results of the COOLFluiD solver and the HDG code.

- $p_0 = 5000\,\mathrm{Pa}$
- $Q = 16\,\mathrm{g\,s^{-1}}$
- $T_{wall} = T_{inlet} = 350\,\mathrm{K}$
- $P = 100\,\mathrm{kW}$
- $f = 0.37\,\mathrm{MHz}$
- air mixture with 11 species.
- No swirl angle.
- The coflow applied for HDG is $\chi_{HDG} = 0.2$.

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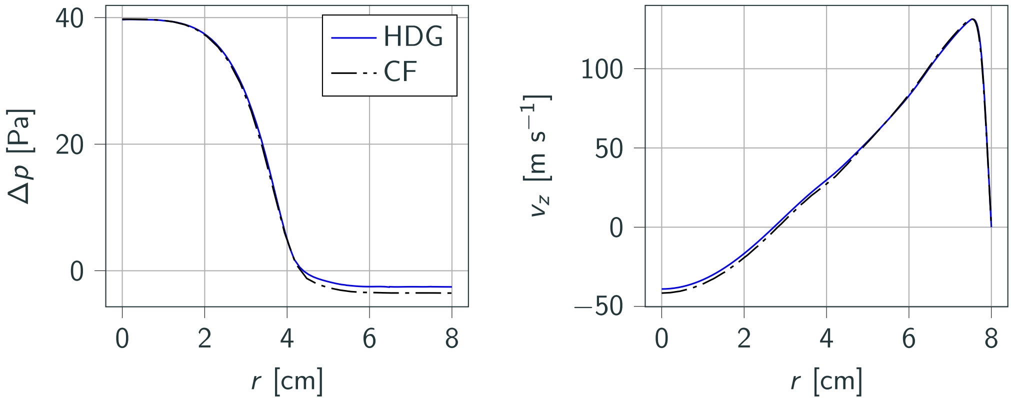

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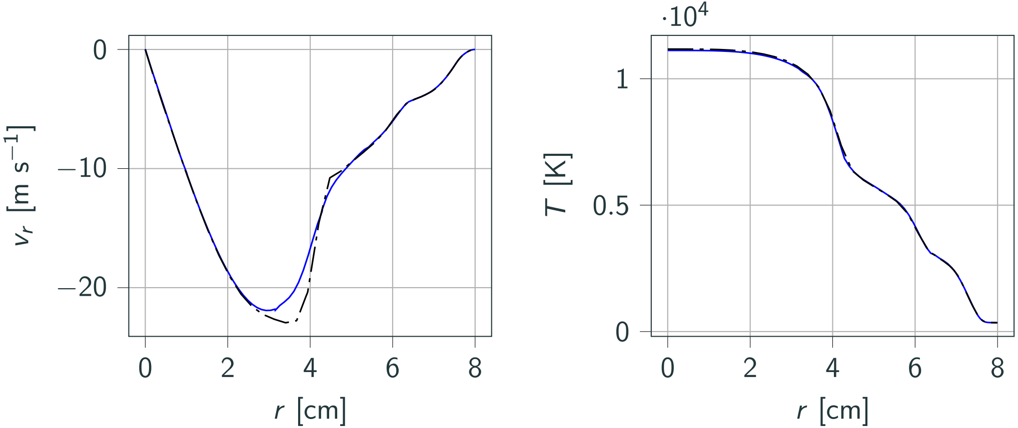

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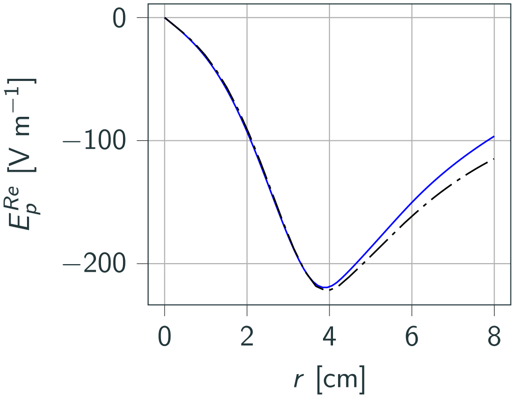
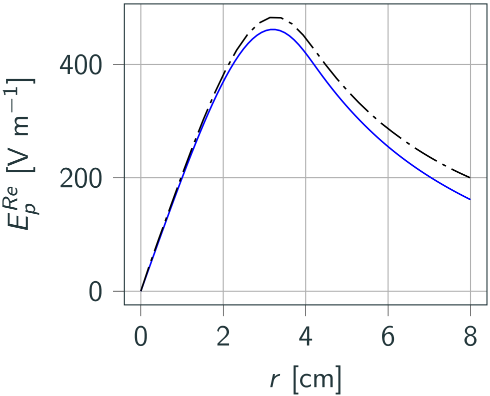




Comparison with the COOLFluiD solver

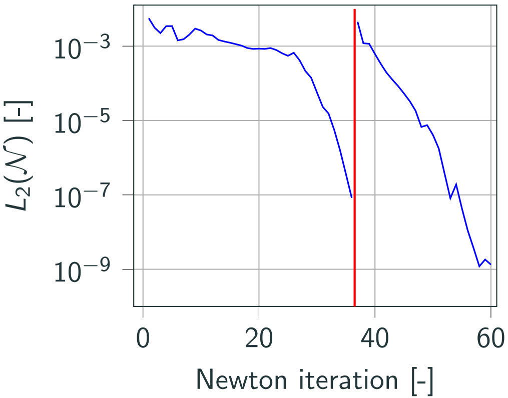
We compared the results of the COOLFluiD solver and the HDG code.

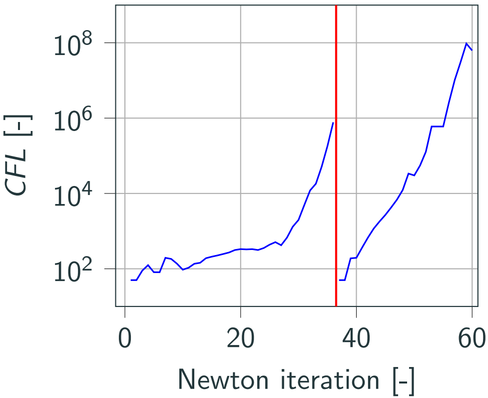
Very similar results except for the electric field in the torch, might be due to the way electric field boundary conditions are imposed.

Convergence history

We use GMRES solver with at most 50 vectors and a ILU(4) preconditioner.

We first solve with polynomial degree 1 then 4 (red separation line).





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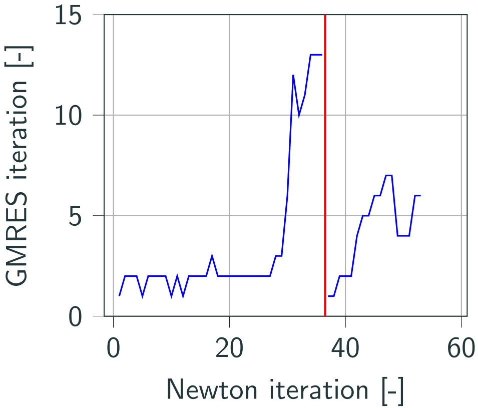

Convergence history

We use GMRES solver with at most 50 **vectors and a ILU(4) preconditioner.**

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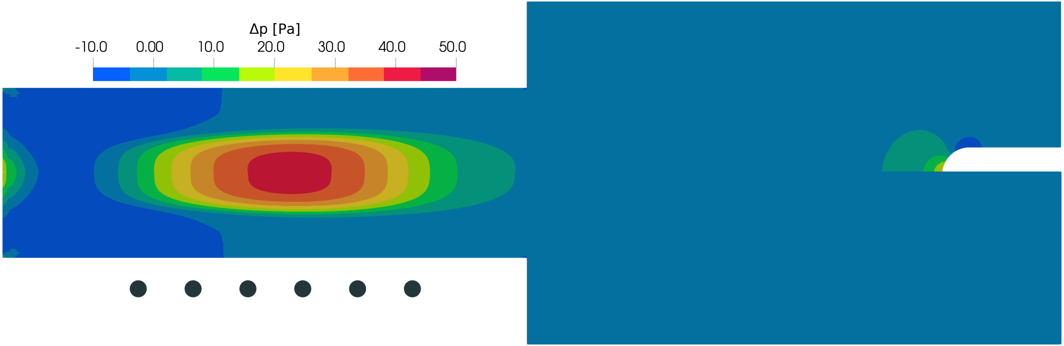
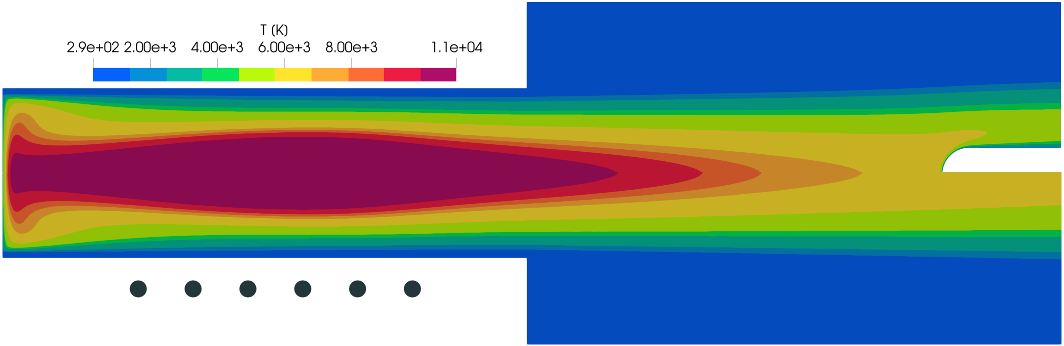
We have a fast and robust solver for ICP, able to produce meaningful results in 20 minutes of time on a single laptop using 8 OMP threads!

Flow fields in ICP: gauge pressure



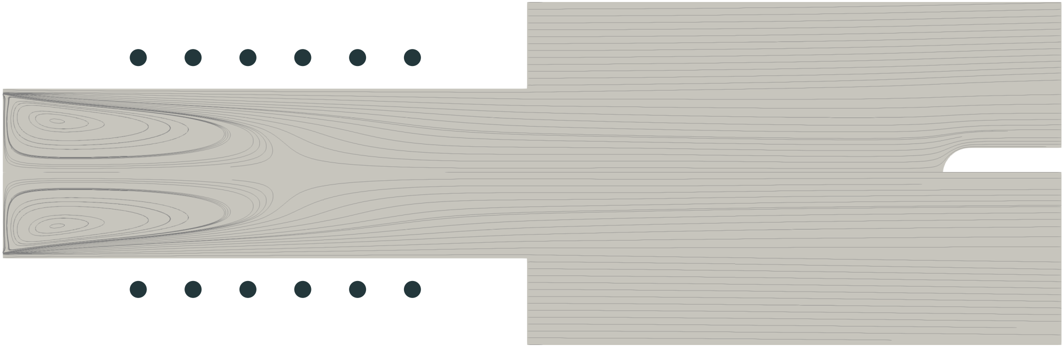
The pressure is almost constant everywhere.

Flow fields in ICP: temperature



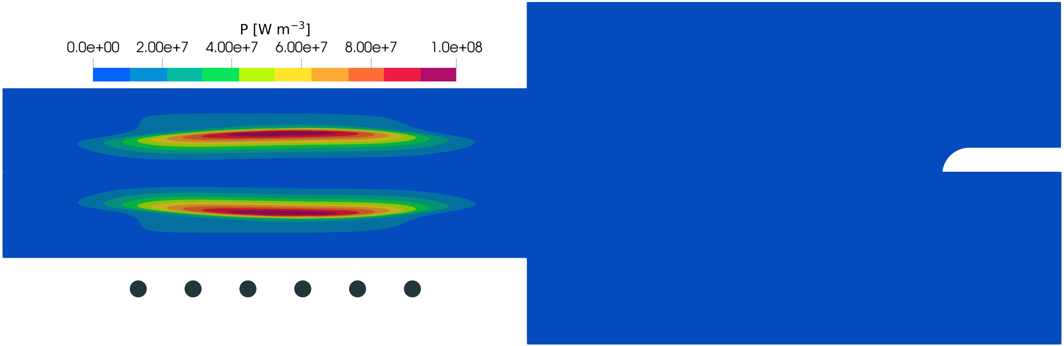
Large temperature gradients.

Flow fields in ICP: streamlines



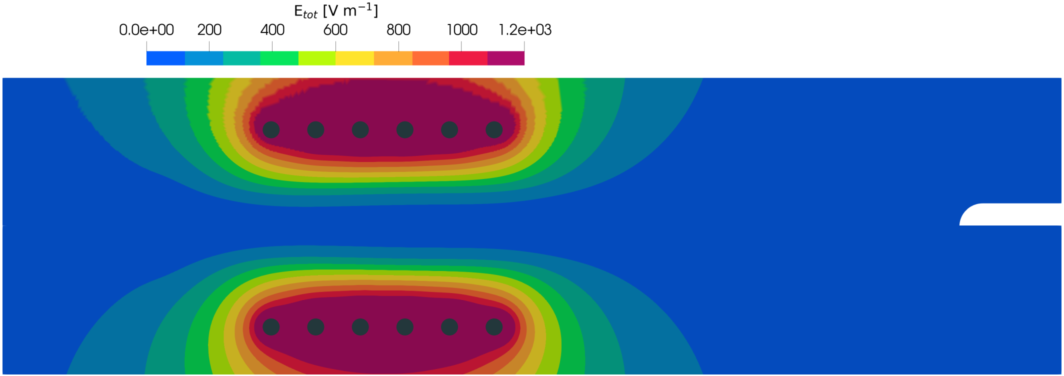
$$\nabla \cdot \mathbf{v} \simeq 0$$

Flow fields in ICP: volume power

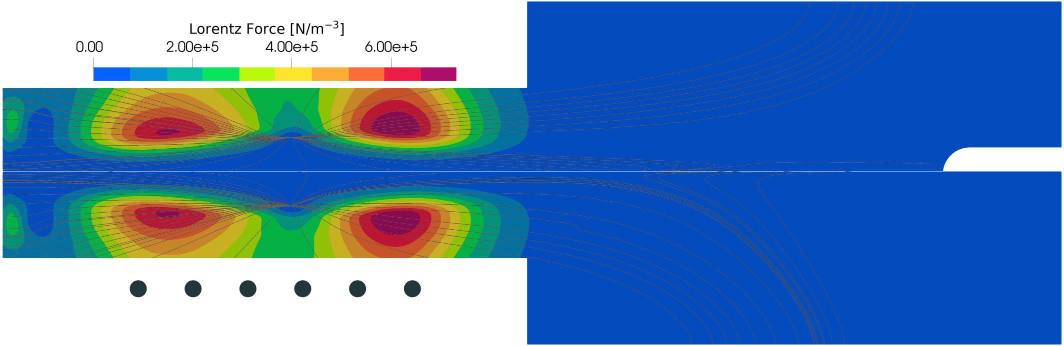


Volume power peak off-axis, where the coupling is the strongest.

Flow fields in ICP: Total electric field



Flow fields in ICP: Lorentz force



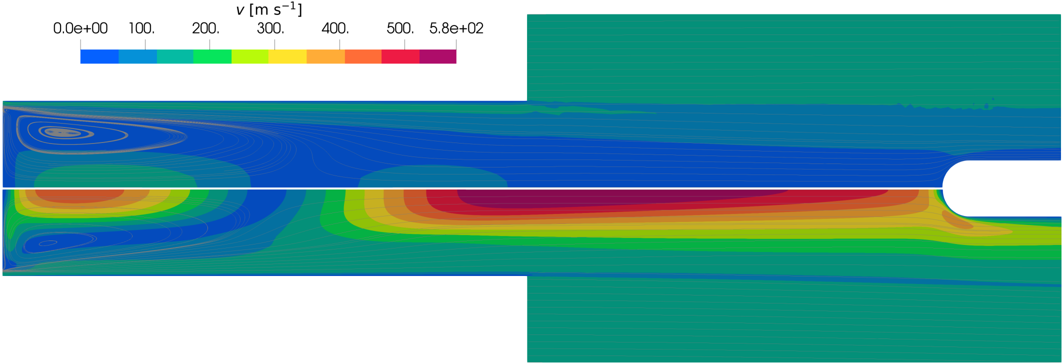
Parametric study

We performed a parametric study of ICP for the parameters used by the experimenters.

- Q from $8\,\mathrm{g\,s^{-1}}$ to $16\,\mathrm{g\,s^{-1}}$.
- p_0 from $5000\,\mathrm{Pa}$ to $20\,000\,\mathrm{Pa}$.
- P_{tot} from $50\,\mathrm{kW}$ to $200\,\mathrm{kW}$.
- f from $10\,\mathrm{kHz}$, to $1\,\mathrm{MHz}$.
- S from 0° to 20°.

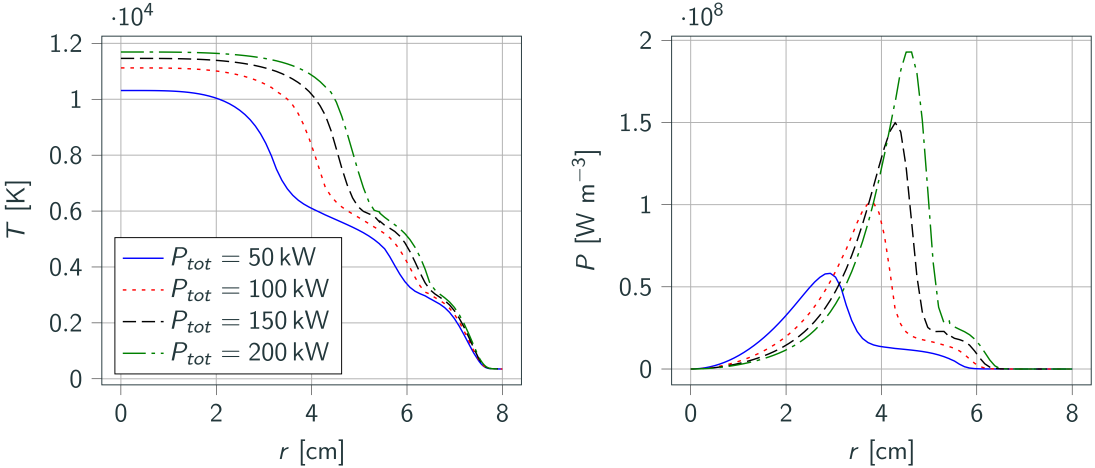
The complete discussion is in the thesis.

Power dissipated in the facility: impact on mass flow rate

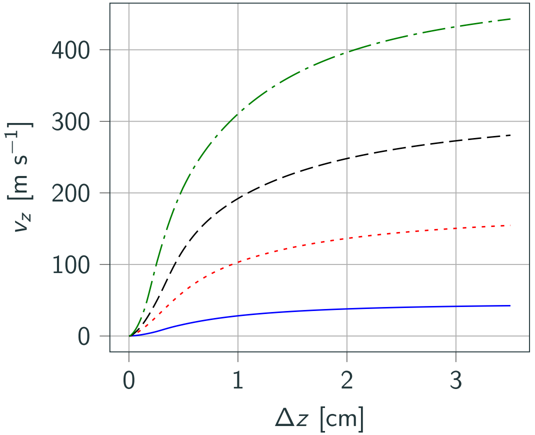


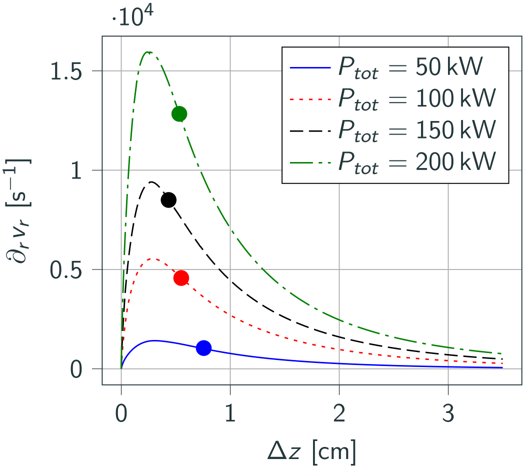
$P_{tot} = 50\,\text{kW}$ (top) and $P_{tot} = 200\,\text{kW}$ (bottom)

Power dissipated in the facility: impact in the torch



Power dissipated in the facility: impact along the stagnation line





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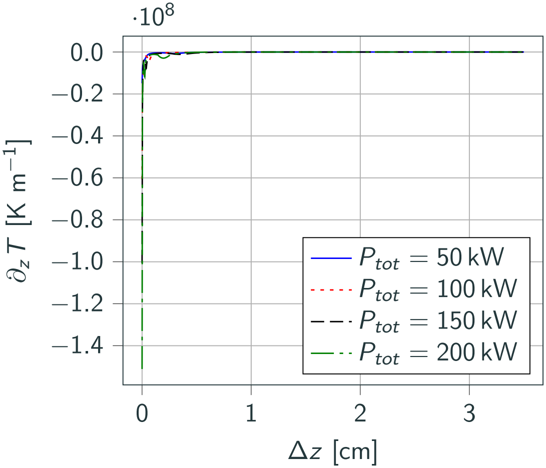

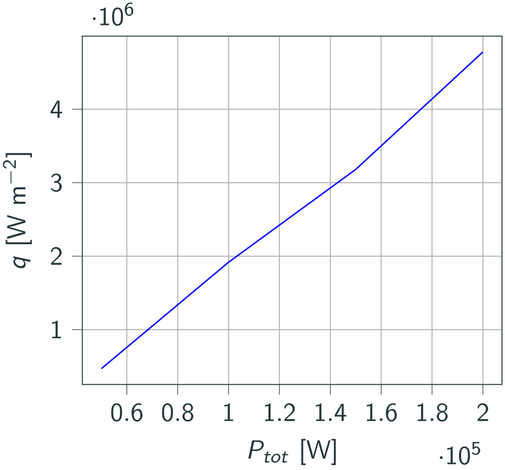

Conclusions and future work

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Q1 In addition of being precise, can a high-order solver be robust?

YES! More robust than FV, thanks to the monolithic strategy.

Q2 Is the developed solver user-friendly?

Yes, but there is still some parameter tuning.

However, much easier to use and give meaningful results fast.

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Conclusions

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Q3 Can the new solver be easily adapted to the new experiments performed in ICP facilities?

We believe so, but there is still much work to do...

Going 2D, 3D, and complex chemistry and radiation for unsteady simulations will need much more computational power.

For supersonic, shock capturing and redesign of the numerical flux.

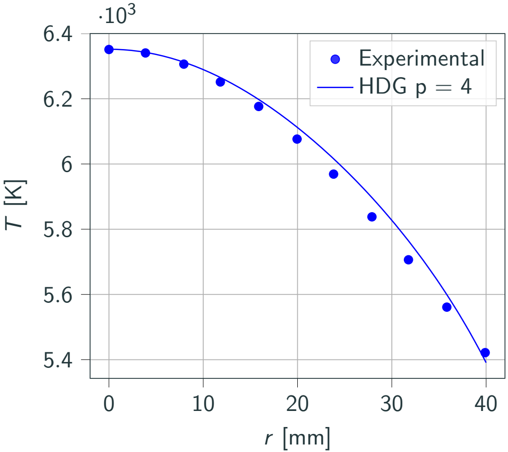
Future work

Main effort is to put our developments in the ARGO DG solver, which is fully paralleled and equipped shock capturing technologies. (Landry Riou)

We see the following roadmap:

- Going 3D.
- Going unsteady (also relieving the average over the induction current period).
- Development of chemistry (non equilibrium).
- Development of radiation.
- For all of the above, comparison with experiment when possible.

Comparison with experiment



- $p_0 = 200\,\text{mbar}$
- $Q = 16\,\text{g s}^{-1}$
- $P_{exp.} = 200\,\text{kW}$
- $P_{num.} = 78\,\text{kW}$
- Freestream, no swirl, air.

Comparison of the experimental $T(r)$ at $385\,\text{mm}$ from the torch exit, from $r = 0$ to $40\,\text{mm}$.